Problem Statement:

Building end-to-end model to detect and transcribe historical Spanish printed text with high accuracy.

To address this problem, a **three-stage OCR pipeline** has been implemented.The stages are as follows:

1. DBNet for text detection
2. CRNN for text recognition.
3. LLM for output refining.

In [1]:
```
# Importing necessary libraries:
import sys
import fitz
import os
import matplotlib.pyplot as plt
import matplotlib.patches as patches
import cv2
from tqdm import tqdm
from doctr.io import DocumentFile
import re
import random
import albumentations as A
import torch
import numpy as np
from torch.utils.data import Dataset
from torch.utils.data import DataLoader
import torch.nn as nn
from IPython.display import FileLink
import ollama
import time
```

```
C:\Users\91861\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5n2kfra8p0\LocalCache\local-packages\Python311\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

Stage 1- Text Detection:

Step 1- Converting image files from pdf to jpg:

In [4]:
```
#Listing paths of the saved pdfs on my local device.
path1=r"C:\Users\91861\Downloads\Print_Test_Files\Print\PORCONES.23.5 - 1628.pdf"
path2=r"C:\Users\91861\Downloads\Print_Test_Files\Print\PORCONES.748.6 – 1650.pdf"
path3=r"C:\Users\91861\Downloads\Print_Test_Files\Print\Guardiola - Tratado nobleza
path4=r"C:\Users\91861\Downloads\Print_Test_Files\Print\PORCONES.228.38 – 1646.pdf"
path5=r"C:\Users\91861\Downloads\Print_Test_Files\Print\Covarrubias - Tesoro lengua
```

```
path6=r"C:\Users\91861\Downloads\Print_Test_Files\Print\Buendia - Instruccion.pdf"
path_list=[path1,path2,path3,path4,path5,path6]
```

In [5]:
```
for i in range(len(path_list)):
    doc=fitz.open(path_list[i])
    print(doc.page_count) #Printing no of pages for each Pdf (sanity check)
```

```
12
26
311
23
990
33
```

In [7]:
```
#Code snippet to open a pdf from it's storage path and read it using the fitz modul
doc1=fitz.open(path1)
doc1.page_count
```

Out[7]: 12

In [12]:
```
base_output_path=r"C:\Users\91861\Desktop\GSOC 26\Human_AI\RenAIssance\Printed_Text
docu_list=["pdf1","pdf2","pdf3","pdf4","pdf5","pdf6"]
```

In [80]:
```
#NO NEED TO RERUN- Jpegs already saved
#Converting all pdfs into jpegs and storing then in respective folders:

for i in range(len(path_list)):
    folder_path = os.path.join(base_output_path, docu_list[i])
    os.makedirs(folder_path, exist_ok=True)
    doc=fitz.open(path_list[i])
    pg_no=doc.page_count

    for j in range(pg_no):
        page=doc.load_page(j)
        pix=page.get_pixmap(dpi=100)
        output_path = os.path.join(folder_path, f"page-{j+1}.jpg")
        pix.save(output_path, jpg_quality=90)

    doc.close()
```

Out[80]:
```
'\nfor i in range(len(path_list)):\n    folder_path = os.path.join(base_output_pat
h, docu_list[i])\n    os.makedirs(folder_path, exist_ok=True)\n    doc=fitz.open(p
ath_list[i])\n    pg_no=doc.page_count\n    \n    for j in range(pg_no):\n
page=doc.load_page(j)\n        pix=page.get_pixmap(dpi=100)\n        output_path =
os.path.join(folder_path, f"page-{j+1}.jpg")\n        pix.save(output_path, jpg_qu
ality=90)\n        \n    doc.close()  '
```

The above code converts the pdfs into jpg files.Now, the next step is text detection on individual page images.

Ground Truth Labels are available for the following page numbers:

1. pdf1- *PORCONES.23.5* - 1628-Pg 1,2,3,4
2. pdf2- *PORCONES.748.6* – 1650-Pg 1,2,3,4

3. pdf3- *Guardiola - Tratado nobleza* - Pg 12,13,14
4. pdf4- *PORCONES.228.38 – 1646* - Pg 1,2,3,4,5
5. pdf5- *Covarrubias - Tesoro lengua* - Pg 7,8,9
6. pdf6- *Buendia - Instruccion* - pg2,3,4

In [81]:
```
#Storing Page nos for which ground truth labels available for each Pdf as a list:

pages_to_process=[[x for x in range(1,5)],[x for x in range(1,5)],[x for x in range
print(pages_to_process)

pdf_gt_page_counts=["4","4","3","5","3","3"]
```

```
[[1, 2, 3, 4], [1, 2, 3, 4], [12, 13, 14], [1, 2, 3, 4, 5], [7, 8, 9], [2, 3, 4]]
```

Step 2- Text Detection:

Model Used: Pre-Trained DBNet Model

Basic Idea : DBNet approaches scene text detection as a pixel-wise segmentation (into text/ not text) problem. Given an input image, a convolutional backbone (e.g., ResNet) is used to extract deep feature maps. A segmentation head is then used to predict a probability map where each value corrosponding to a pixel represents the likelihood of the belonging to a text region.

Instead of using a fixed threshold (like more traditional approaches) to convert this probability map into a binary mask, DBNet introduces the concept of a learnable threshold map.This allows the model to adaptively determine decision boundaries at each pixel location.

Differrentiable Binarization: Traditional binarization applies a hard threshold-

$$B(x,y) = \begin{cases} 1 & \text{if } P(x,y) > T(x,y) \\ 0 & \text{otherwise} \end{cases}$$

However, this makes it non-differrentiable and thus non-trainable.

The innovation of the DBNet Model is the use of the following approximation:

$$\hat{B}(x,y) = \sigma\big(k(P(x,y) - T(x,y))\big)$$

This approximates hard thresholding while remaining differentiable, thus enabling gradients to flow through both the probability and threshold maps during training. This the complete model can be trained together.

Reasons for Choosing DBNet:

1. Works well with multi-oriented and curved text- In our use case, text may be non-horizontal/curved and handling these cases is crucial.

2. It shows strong performance on the standard scene text benchmarks.

Reference :

Liao, M.; Wan, Z.; Yao, C.; Chen, K.; & Bai, X. (2019).
Real-time Scene Text Detection with Differentiable Binarization.
https://arxiv.org/abs/1911.08947

Implementing DBNet Model:

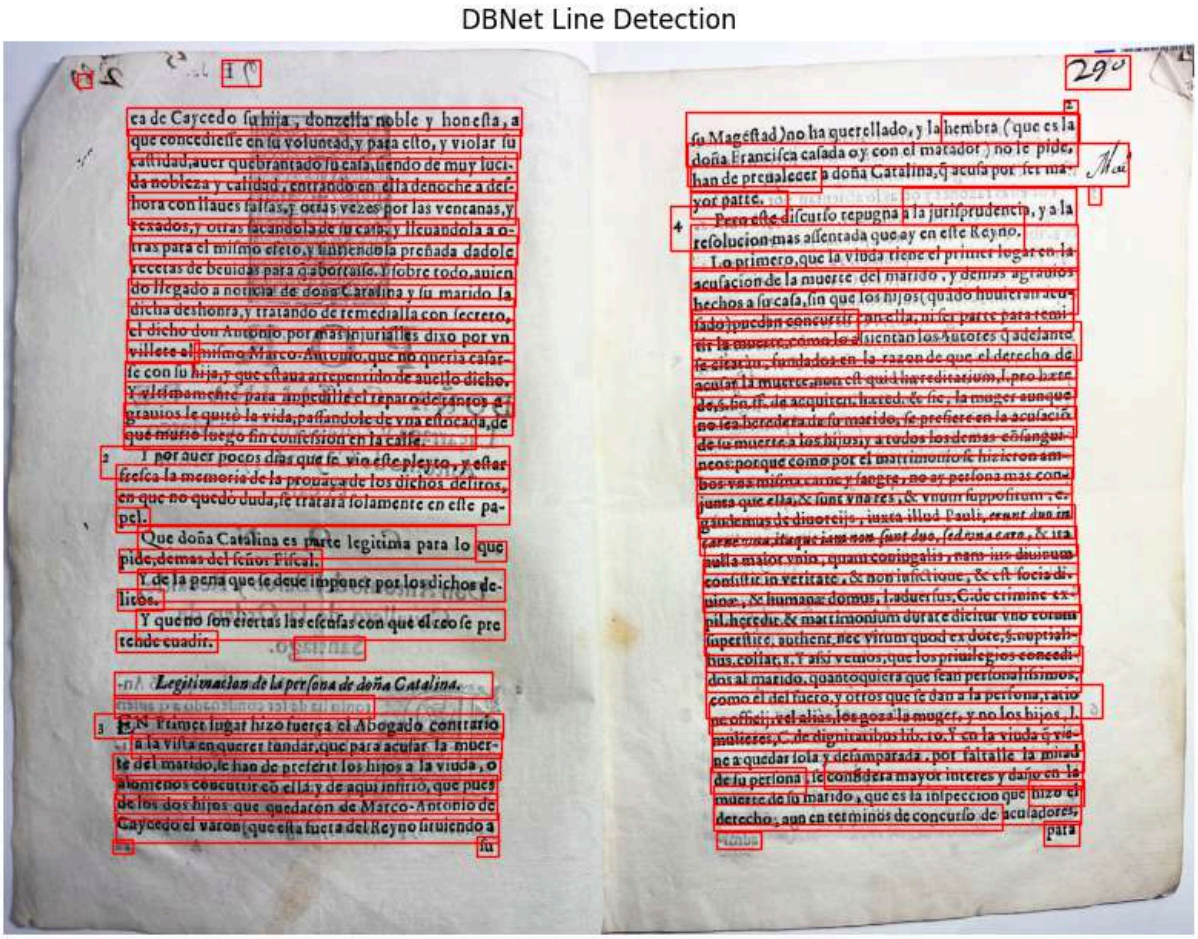
In [125…
```
#Library used- DocTR:

from doctr.models import ocr_predictor #Importing pre-trained OCR model- only text

model = ocr_predictor(pretrained=True) #loading a pretrained OCR pipeline

#First, implementing text detection on single page:
#path of Page 2 of Pdf1 saved on my system:
path=r"C:\Users\91861\Desktop\GSOC 26\Human_AI\RenAIssance\Printed_Text\pdf1\page-2

doc = DocumentFile.from_images(path) #Converts image into DocTR's internal format-
result = model(doc) #Applying OCR to the doc object
```

In [46]:
```
#Printing coordinates of the first 4 detected bounding boxes:

page = result.pages[0] #Accessing 1st page in the result.pages list- here, it conta
for block in page.blocks: #Block= group of lines
    i=1
    for line in block.lines:
        while i<5:
            print(line.geometry) #line.geometry gives the bounding boxes of each li
            i=i+1           #coordinates are normalized between 0 and 1- need to be
```

```
((np.float64(0.0615234375), np.float64(0.036458333333333315)), (np.float64(0.0732421
875), np.float64(0.052083333333333315)))
((np.float64(0.0615234375), np.float64(0.036458333333333315)), (np.float64(0.0732421
875), np.float64(0.052083333333333315)))
((np.float64(0.0615234375), np.float64(0.036458333333333315)), (np.float64(0.0732421
875), np.float64(0.052083333333333315)))
((np.float64(0.0615234375), np.float64(0.036458333333333315)), (np.float64(0.0732421
875), np.float64(0.052083333333333315)))
```

In [128…
```
#Visualizing the detected bounding boxes on the Printed page image:

fig, ax = plt.subplots(figsize=(10, 15))
img = doc[0]
ax.imshow(img)

h, w = img.shape[:2]

#Drawing bounding boxes using the coordinates in the image:
for block in page.blocks:
    for line in block.lines:
        (x_min, y_min), (x_max, y_max) = line.geometry
```

```
        #calculating positions in the image from the normalized values using image
        x_min *= w
        x_max *= w
        y_min *= h
        y_max *= h

        rect = patches.Rectangle(
            (x_min, y_min),# top left coordinates of bounding box- anchor point
            x_max - x_min,# width = Right boundary-Left boundary
            y_max - y_min,# height = Bottom boundary-Top boundary
            linewidth=1,
            edgecolor='red',
            facecolor='none'
        ) #creating the rectangular patch object- the bounding box

        ax.add_patch(rect) #adding the bounded box to the image

plt.title("DBNet Line Detection")
plt.axis('off')
plt.show()
```



Step 3- Dataset Preparation:

The model is able to satisfactory detect lines in the text as seen from the above image.
So, we move forward with cropping the image following the bounding boxes and storing the

individual cropped line images.These images will be the inputs to be used for model training.

In [31]:
```
#Base directory for saving the line crops of Pdf1, pg2:

base_dir = r"C:\Users\91861\Desktop\GSOC 26\Human_AI\RenAIssance\Printed_Text\pdf1\
os.makedirs(base_dir, exist_ok=True)

img = doc[0]
h, w, _ = img.shape #Derieving image height and width

page = result.pages[0] #Derieving the results of OCR on the Page

line_id = 0

for block in page.blocks:
    for line in block.lines:

        (x_min, y_min), (x_max, y_max) = line.geometry

        x_min = int(x_min * w)
        x_max = int(x_max * w)
        y_min = int(y_min * h)
        y_max = int(y_max * h)

        crop = img[y_min:y_max, x_min:x_max]
        save_path = os.path.join(base_dir, f"line_{line_id:04d}.jpg")

        cv2.imwrite(save_path, cv2.cvtColor(crop, cv2.COLOR_RGB2BGR))

        line_id += 1

print(line_id) #This is the number of lines extracted from the image.
```

```
Total lines extracted: 85
```

Next,for all pages of the Pdfs which have corrosponding ground truth labels, we repeat the process of generating bounding boxes, cropping the lines following the bounding boxes and saving the cropped images.

In [57]:
```
#NO NEED TO RERUN - Cropped line images already stored.

# Flattening total page count for a progress bar:
total_pages = sum(len(pages) for pages in pages_to_process)

#Setting base path:
base_root = r"C:\Users\91861\Desktop\GSOC 26\Human_AI\RenAIssance\Printed_Text"

not_done_log=[]

with tqdm(total=total_pages, desc="Processing Pages") as pbar:

    for pdf_index in range(6):
```

```
        pdf_name = f"pdf{pdf_index + 1}"
        pdf_folder = os.path.join(base_root, pdf_name)

        main_crop_folder = os.path.join(pdf_folder, "line_crops")
        os.makedirs(main_crop_folder, exist_ok=True)

        for page_number in pages_to_process[pdf_index]:

            image_path = os.path.join(pdf_folder, f"page-{page_number}.jpg")

            if not os.path.exists(image_path):
                pbar.update(1)
                not_done_log.append(f"Pdf: {pdf_name} , Pg No: {page_number} could
                continue

            doc = DocumentFile.from_images(image_path)
            result = model(doc)

            img = doc[0]
            h, w, _ = img.shape
            page = result.pages[0]

            page_folder = os.path.join(main_crop_folder, f"page_{page_number}")
            os.makedirs(page_folder, exist_ok=True)

            line_id = 0

            for block in page.blocks:
                for line in block.lines:

                    (x_min, y_min), (x_max, y_max) = line.geometry

                    x_min = int(x_min * w)
                    x_max = int(x_max * w)
                    y_min = int(y_min * h)
                    y_max = int(y_max * h)

                    if x_max > x_min and y_max > y_min:

                        crop = img[y_min:y_max, x_min:x_max]

                        save_path = os.path.join(
                            page_folder,
                            f"line_{line_id:04d}.jpg")

                        cv2.imwrite(
                            save_path,
                            cv2.cvtColor(crop, cv2.COLOR_RGB2BGR))

                        line_id += 1

            pbar.update(1)
```

```
rocessing Pages: 100%|████████████████████████████████████████████████████████
██| 22/22 [02:20<00:00,  6.41s/it]
```

Preparing the ground truth labels (The Y variable values) for model training:

In [1]:
```
#NO NEED TO RERUN - Cropped line image labels already stored.

# Base path where all Ground Truth labels are stored as .txt files:
gt_folder = r"C:\Users\91861\Downloads"

#File naming structure of the ground truth text files - gt_i_j, where i=pdf no and

# Dictionary structure is used to store the gt text- nested dictionary
# ground_truth[pdf_no][page_no] = gt_text

ground_truth = {}

pattern = re.compile(r"gt_(\d+)_(\d+)\.txt")

for filename in os.listdir(gt_folder):
    match = pattern.match(filename)

    if match:
        pdf_no = int(match.group(1))
        page_no = int(match.group(2))
        file_path = os.path.join(gt_folder, filename)

        with open(file_path, "r", encoding="utf-8") as f:
            text = f.read().strip()

        if pdf_no not in ground_truth:
            ground_truth[pdf_no] = {}

        ground_truth[pdf_no][page_no] = text
print("Ground truths loaded successfully!")
```

```
Ground truths loaded successfully!
```

The input images consist of line level data.So, we split the ground truth text for each page line by line.

In [4]:
```
#NO NEED TO RERUN - Cropped line image labels already stored.

#The input images consist of line level data.So, we split the ground truth text for
for pdf_no in ground_truth:
    for page_no in ground_truth[pdf_no]:

        full_text = ground_truth[pdf_no][page_no]

        # Splitting by line breaks and removing empty lines
        lines = [line.strip() for line in full_text.splitlines() if line.strip() !=

        ground_truth[pdf_no][page_no] = lines

#This will convert the transcripts into line level lists.
```

In [6]:
```
ground_truth[1][2][:10]    #Sanity Check
```

Out[6]:
```
['ca de Caycedo su hija, donzella noble y honesta, a',
 'que concedielle en su voluntad, y para esto, y violar su',
 'castidad, aver quebrantado su casa, siendo de muy luci-',
 'da nobleza y calidad, entrando en ella denoche a des-',
 'hora con llaves falsas, y otras vezes por las ventanas, y',
 'texados, y otras sacandola de su casa, y llevandola a o-',
 'tras para el mismo efeto, y sintiendola preñada dadole',
 'recetas de bevidas para q abortasse. Y sobre todo, avien',
 'do llegado a noticia de doña Catalina y su marido la',
 'dicha deshonra, y tratando de remedialla con secreto,']
```

In [7]:
```
#NO NEED TO RERUN - Cropped line image labels already stored.

#Generating a file "labels" for each page that maps the cropped line images(ordered

base_root = r"C:\Users\91861\Desktop\GSOC 26\Human_AI\RenAIssance\Printed_Text"

for pdf_no in ground_truth:
    pdf_folder = os.path.join(base_root, f"pdf{pdf_no}", "line_crops")
    if not os.path.exists(pdf_folder):
        continue

    for page_no in ground_truth[pdf_no]:
        page_folder = os.path.join(pdf_folder, f"page_{page_no}")
        if not os.path.exists(page_folder):
            continue

        image_files = [
            f for f in os.listdir(page_folder)
            if f.lower().endswith(".jpg")]

        image_files.sort()
        gt_lines = ground_truth[pdf_no][page_no]
        max_len = max(len(image_files), len(gt_lines))
        labels_path = os.path.join(page_folder, "labels.txt")

        with open(labels_path, "w", encoding="utf-8") as f:
            for i in range(max_len):
                img_name = image_files[i] if i < len(image_files) else ""
                gt_line = gt_lines[i] if i < len(gt_lines) else ""
                f.write(f"{img_name}\t{gt_line}\n")
```

Note: This saves the images of text lines and line labels in files.I have manually removed some blank images and rearranged some files to form correct the (Cropped Line Image, ground Truth Line Text) pair mapping, for a few cases.
Thus we have a dataset with **(Cropped Line Image, ground Truth Line Text)** pairs ready for training.

Stage 2- CRNN Modelling:

Step 4- Preparing Train, Validation, Test Data:

Loading the complete dataset and shuffling it before splitting into Train- Validation- Test sets.

Dataset Shuffling is done to prevent lines from only one/two pdfs appearing in the Validation/Test set and thus leading to incorrect accuracy metrics.

In [2]:
```
# Root directory containing the pdfs:
root_dir = r"C:\Users\91861\Desktop\GSOC 26\Human_AI\RenAIssance\Printed_Text"

dataset = []

for pdf_folder in os.listdir(root_dir):
    pdf_path = os.path.join(root_dir, pdf_folder)
    if not os.path.isdir(pdf_path):
        continue

    line_crops_path = os.path.join(pdf_path, "line_crops")
    if not os.path.exists(line_crops_path):
        continue

    for page_folder in os.listdir(line_crops_path):
        page_path = os.path.join(line_crops_path, page_folder)
        if not os.path.isdir(page_path):
            continue
        labels_file = os.path.join(page_path, "labels.txt")

        if not os.path.exists(labels_file):
            continue

        with open(labels_file, "r", encoding="utf-8") as f:
            for line in f:
                line = line.strip()

                if not line:
                    continue

                img_name, text_label = line.split("\t")
                img_path = os.path.join(page_path, img_name)
                dataset.append((img_path, text_label))


# shuffling dataset:
random.seed(42)
random.shuffle(dataset)

# Sanity Check:
print("Total dataset size:", len(dataset))
print("Last 5 data points", dataset[:5])
```

```
Total dataset size: 816
Last 5 data points [('C:\\Users\\91861\\Desktop\\GSOC 26\\Human_AI\\RenAIssance\\Pri
nted_Text\\pdf6\\line_crops\\page_3\\line_0023.jpg', 'dad.'), ('C:\\Users\\91861\\De
sktop\\GSOC 26\\Human_AI\\RenAIssance\\Printed_Text\\pdf4\\line_crops\\page_4\\line_
0029.jpg', 'mez Pimentel, ha recibido de Antonio Gomez su'), ('C:\\Users\\91861\\Des
ktop\\GSOC 26\\Human_AI\\RenAIssance\\Printed_Text\\pdf4\\line_crops\\page_3\\line_0
012.jpg', 'traslado dellas a las partes, y que Antonio Gomez'), ('C:\\Users\\91861
\\Desktop\\GSOC 26\\Human_AI\\RenAIssance\\Printed_Text\\pdf5\\line_crops\\page_8\\l
ine_0038.jpg', 'salieron deste Colegio, de donde avia salido aquel varon nunca acaba
do de alabar, el Doctor'), ('C:\\Users\\91861\\Desktop\\GSOC 26\\Human_AI\\RenAIssan
ce\\Printed_Text\\pdf6\\line_crops\\page_4\\line_0010.jpg', 'gel, y Barcelona, O
c.')]
```

Train- Validation- Test split:

I am training on 90% of the dataset (As the dataset is very small, I am keeping as much data possible for better training) and allocating 5% each for Validation and Test Datasets.

In [3]:
```python
from sklearn.model_selection import train_test_split

# First splitting into: train vs (test&val)
train_data, test_and_val_data = train_test_split(dataset,test_size=0.10,random_stat

# Second split: validation vs test
val_data, test_data = train_test_split(test_and_val_data,test_size=0.5,random_state

# Printing dataset sizes
print("Train set size:", len(train_data))
print("Validation set size:", len(val_data))
print("Test set size:", len(test_data))
```

```
Train set size: 734
Validation set size: 41
Test set size: 41
```

Step 5: Carrying out data augmentation on Training Data:

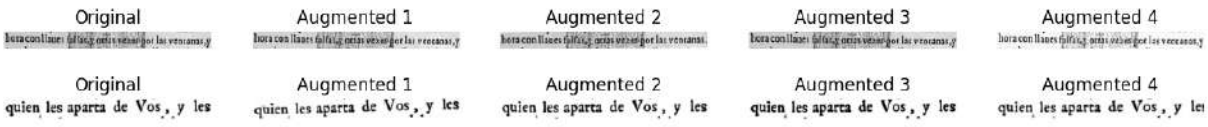
Data augmentation prevents overfitting- It helps learn more robust features.

In [4]:
```python
#Creating a data augmentation pipeine:
train_transform = A.Compose([
    # 1. Image Rotation:
    A.Rotate(limit=2,#Randomly rotates the image--2° to +2°.
        border_mode=cv2.BORDER_CONSTANT,fill=255,#rotation creates empty space arou
        p=0.4), # 40% of images get rotated,

    # 2. Shifting and resizing image:
    A.ShiftScaleRotate(shift_limit=0.01, #shifting image vertically/horizontally- m
        scale_limit=0.05, #scaling image a max of 5% up or down
        rotate_limit=0,border_mode=cv2.BORDER_CONSTANT,fill=255,p=0.4),

    # 3. Adjuting brightness/contrast:
    A.RandomBrightnessContrast(brightness_limit=0.2, #inc/ dec brigtness by max of
        contrast_limit=0.2, #inc/ dec contrast by max of 20%
```

```
        p=0.5 #randomly applied to 1/2 of the images
    ),

    # 4. Adding Gaussian blur:
    A.GaussianBlur(blur_limit=(3,3),p=0.3), # blur applied on 30% of images.

    # 5. Adding Gaussian noise:
    A.GaussNoise(std_range=(0.02, 0.08),p=0.3), #adding random pixel noise.

    # 6. Adding slight elastic distortion (warping of the image).
    A.ElasticTransform(alpha=1,sigma=30,border_mode=cv2.BORDER_CONSTANT,fill=255,p=
```

```
C:\Users\91861\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5n2kfra8p0\LocalCache\local-packages\Python311\site-packages\albumentations\core\validation.py:114: UserWarning: ShiftScaleRotate is a special case of Affine transform. Please use Affine transform instead.
  original_init(self, **validated_kwargs)
```

In [10]:
```
#Visualising the data augmentation results on 2 data points:

# picking first two images from the training dataset:
image_paths = [train_data[0][0], train_data[1][0]]

for path in image_paths:
    img = cv2.imread(path, cv2.IMREAD_GRAYSCALE) #Loading image as grayscale.
    plt.figure(figsize=(15,3))

    plt.subplot(1,5,1)
    plt.imshow(img, cmap="gray")
    plt.title("Original")
    plt.axis("off")

    for i in range(4):
        augmented = train_transform(image=img)["image"] #generating augmentations
        plt.subplot(1,5,i+2)
        plt.imshow(augmented, cmap="gray")
        plt.title(f"Augmented {i+1}")
        plt.axis("off")

    plt.show()
```



The augmented images look realistic, so we are going to apply this pipeline to all images while training so that our model is trained on a larger and more varied dataset.

Preparing Character Set:

CRNN Model carries out sequence prediction where each character predicted is a classification over all possible characters.So, first, we must define the Character Set.

The final softmax layer in the Model will have (Character Set size +1[For Blank]) neurons- each neuron will corrospond to a character.

In [11]:
```
# Extracting character set from labels:
charset = set()

for _, label in dataset:
    for character in label:
        charset.add(character)

charset = sorted(list(charset))

print(charset)
print("Length of Character Set: ", len(charset))
```

```
[' ', '!', '&', '(', ')', ',', '-', '.', '0', '1', '2', '3', '4', '5', '6', '7',
'8', '9', ':', ';', '?', 'A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'L', 'M',
'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'Y', 'Z', 'a', 'b', 'c', 'd', 'e', 'f',
'g', 'h', 'i', 'j', 'l', 'm', 'n', 'o', 'p', 'q', 'r', 's', 't', 'u', 'v', 'x', 'y',
'z', 'Ş', 'Ñ', 'á', 'ç', 'é', 'í', 'ñ', 'ó', '–', '…']
Length of Character Set:  78
```

Thus, the last layer of the CRNN model will have 79 Neurons.

Theory and Reasoning Behind Models/Frameworks Used:

CRNN Model:

Basic Idea:
CRNN is a combination of:

1. DeepCNN
2. RNN(Deep Bi-Directional LSTM)
3. CTC Transcription layer

Motivation: Why is traditional CNN not suitable for this task?

1. CNNs are effective for image classification tasks where the goal is to predict a single label for an input image. However, image-based text recognition requires prediction of a **sequence of labels** rather than a single label.
2. CNN architecture typically produces fixed output dimensions. In OCR tasks, the length of the target output sequence can vary significantly.
   Thus, standard CNN architecture alone is not well suited for text recognition tasks.

Why is RNN Used?

1. Recurrent Neural Networks are specifically designed to **model sequential data**. In image-based text recognition, characters appear sequentially along the horizontal

direction of the image which can be modelled with RNN.
2. Characters can often be recognized more reliably when their surrounding context is considered. For example, ambiguous characters such as “i” and “l” can be distinguished more easily when neighboring characters are taken into account. (**Reference** -Shi, Bai & Yao (2015)- *An End-to-End Trainable Neural Network for Image-based Sequence Recognition and Its Application to Scene Text Recognition*)
3. RNNs can process **sequences of variable** length, making them suitable for OCR tasks where different images contain different numbers of characters.

To capture long-range dependencies in text sequences, CRNN uses **Long Short-Term Memory (LSTM) units**, which solves the vanishing gradient problem present in traditional RNNs.

Model Architecture:

1. **Convolutional Layers:**

   -Extracts visual feature maps from the input image.
   -Workflow is: image → convolutional feature maps → Map-to-Sequence transformation → feature sequence
   -Each column in a feature map corresponds to a receptive field in the original image.Thus, the model effectively reads the image left to right.
   -The feature maps are converted into a sequence of feature vectors- by concatenating the corrosponding columns of all feature maps together to form a feature vector. Thus, this converts image data into sequential data.
   -CRNN removes the fully connected layers used in traditional CNNs and uses only convolution and pooling layers.
   -All input images must be normalized to a fixed height.The width is scaled proportionally to preserve the aspect ratio of the image.

2. **Recurrent Layers:**

   -The feature sequence generated from the feature maps are fed into deep bidirectional LSTM layers.
   -Bidirectional LSTM is used to capture context from both directions, improving recognition accuracy. -Multiple bidirectional LSTM layers are stacked to enable the network to learn higher-level sequential representations.

3. **Transcription Layer:**

   -The transcription layer converts the sequence of predictions produced by the LSTM network into the final text sequence.
   -CRNN uses **Connectionist Temporal Classification (CTC)** to compute the probability of label sequences.CTC introduces a special "blank" token and allows repeated labels in intermediate predictions. A mapping function collapses repeated labels and removes

blanks to produce the final output sequence.The final prediction is obtained by selecting the label sequence with the highest probability.
-**NOTE:** CRNN does not require character-level alignment between the input image and the output text.Older OCR models needed character-level labelled datasets.CRNN instead performs sequence recognition directly- this makes training dataset peperation easier.

Model Training:

The model is trained by **minimizing the negative log-likelihood of the predicted label sequence, conditional on the input image**.

Key Advantages:

1. CTC enables the model to learn from image–text pairs directly, without requiring explicit segmentation of characters- thus character level annotations are not required.
2. Entire architecture can be optimized jointly in an end-to-end manner.
3. The model can handle variable length sequences naturally.

Reference: Shi, Bai & Yao (2015)
An End-to-End Trainable Neural Network for Image-based Sequence Recognition and Its Application to Scene Text Recognition
https://arxiv.org/abs/1507.05717

Why am I using pre-trained weights:

The prepared dataset contains a small number of training samples (< 800 line images), which is typically insufficient to train a deep neural network like CRNN from scratch. Therefore, instead of initializing the model with random weights, we start with **pre-trained weights** obtained from models trained on large synthetic OCR datasets. These pretrained models have already learned general visual features of text such as character strokes, edges, and spatial patterns.

By **fine-tuning** the pretrained model on the available dataset, the network can adapt these learned representations to the specific characteristics of the task at hand. This transfer learning approach significantly improves training efficiency, reduces overfitting, and enables good recognition performance even with a relatively small dataset.

The **Deep Text Recognition Benchmark framework** provides several downloadable pre-trained models that can be used as a starting point for training OCR systems. These models have been trained on large synthetic text datasets such as MJSynth (Synth90k) and SynthText, which together contain millions of generated word images rendered using a wide variety of fonts, backgrounds, and distortions.

A **Four-stage Scene Text Recognition framework** was introduced in the benchmark paper :

Baek et.al (2019)
What Is Wrong With Scene Text Recognition Model Comparisons? Dataset and Model Analysis
https://arxiv.org/pdf/1904.01906

The framework the paper described is as follows:
Transformation → FeatureExtraction → SequenceModeling → Prediction

1. What is the **Transformation stage**?
   Transforming the input image X into a normalized image X˜.This stage typically uses a **Thin Plate Spline (TPS) Spatial Transformer Network**, which learns to rectify or normalize the input text image by correcting geometric distortions such as slight curvature, perspective skew, or irregular alignment. In our case, the dataset consists of cropped line images from scanned documents where the text is largely horizontal and well aligned. Therefore, a TPS transformation is not expected to significantly affect recognition performance.

Additionally, a TPS layer is not present in the CRNN architecture described in the Shi, Bai & Yao (2015) paper.

For these two reasons, I am not including a Transformation layer in my model architecture.

3. The **Feature Extraction stage** involves using a CNN model to extract feature maps.In the standard architecture used in the Deep Text Recognition Benchmark, the feature extraction stage is implemented using a **ResNet-based Convolutional Neural Network (CNN)**. ResNet is chosen because its residual connections allow very deep networks to be trained effectively without suffering from vanishing gradients.

4. The **Sequence Modelling stage** involves using a Bi-directional LSTM Model.

5. The **Prediction stage** converts the sequence features produced by the BiLSTM layers into the final character sequence. The prediction approach implemented here is Connectionist Temporal Classification (CTC), following Shi, Bai & Yao (2015) paper.

Final Model- I will be implementing the following framework : **None-ResNet-BiLSTM-CTC** (following Shi, Bai & Yao (2015) paper)

Step 6- Defining Model Architecture:

In [12]:
```
#Model -ResNet-BiLSTM-CTC
#Model code download link- https://github.com/clovaai/deep-text-recognition-benchma

#adding the deep-text-recognition-benchmark folder to Python's search path:
sys.path.append(r"C:\Users\91861\Downloads\deep-text-recognition-benchmark-master\d

#importing Model class from module.py:
from model import Model
```

In [13]:
```
#Model configuration- defining model settings/parameters.
class Opt:
    # Model architecture:
    Transformation = "None" #No transformation module.
    FeatureExtraction = "ResNet"
    SequenceModeling = "BiLSTM"
    Prediction = "CTC"

    # image parameters:
    imgH = 32
    imgW = 1024 #this value is a placeholder as no TPS

    # TPS parameters
    #num_fiducial = 20

    # CNN parameters
    input_channel = 1 #img is greyscale, not RGB
    output_channel = 512 #number of feature maps produced by the CNN

    # LSTM parameters
    hidden_size = 256

    # number of output classes:
    num_class = 37 #Temporarily, for compatibility with the downloaded model weight
    #whose weights I will be using.
    #Later on, will change size of output layer to len(charset)+1 when I will repla
    #(each class corrosdponds to 1 character in my character set + 1 class for CTC

opt = Opt() #Creating instance of Opt class
model = Model(opt)
```

```
No Transformation module specified
```

In [14]:
```
#Model weights download Link- https://drive.google.com/drive/folders/15WPsuPJDCzhp2

# loading checkpoint-containing snapshot of the entire trained model's parameters (
ckpt = torch.load(r"C:\Users\91861\Downloads\TPS-ResNet-BiLSTM-CTC.pth",map_locatio

print(type(ckpt))
```

```
<class 'collections.OrderedDict'>
```

In [15]:
```
#Thus the checkpoint data structure is an ordered dictionary.
#Syntax: {layer.parameter_type: tensor(   ),....}

#Printing some keys of the ordered dictionary:
print(list(ckpt.keys())[:5])
print(list(ckpt.keys())[-5:])
```

```
['module.Transformation.LocalizationNetwork.conv.0.weight', 'module.Transformation.L
ocalizationNetwork.conv.1.weight', 'module.Transformation.LocalizationNetwork.conv.
1.bias', 'module.Transformation.LocalizationNetwork.conv.1.running_mean', 'module.Tr
ansformation.LocalizationNetwork.conv.1.running_var']
['module.SequenceModeling.1.rnn.bias_hh_l0_reverse', 'module.SequenceModeling.1.line
ar.weight', 'module.SequenceModeling.1.linear.bias', 'module.Prediction.weight', 'mo
dule.Prediction.bias']
```

In [11]:
```
#The keys corrosponding to the 4 stages in the Framework:
#Transformation → TPS (Which I will not need)
#FeatureExtraction → ResNet
#SequenceModeling → BiLSTM
#Prediction → final classifier
```

In [16]:
```
print(model) #As we can see, model has ResNet, Bidirectional LSTM and Prediction la
#TPS layer is skipped.
```

```
Model(
  (FeatureExtraction): ResNet_FeatureExtractor(
    (ConvNet): ResNet(
      (conv0_1): Conv2d(1, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bi
as=False)
      (bn0_1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_s
tats=True)
      (conv0_2): Conv2d(32, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), b
ias=False)
      (bn0_2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_s
tats=True)
      (relu): ReLU(inplace=True)
      (maxpool1): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mod
e=False)
      (layer1): Sequential(
        (0): BasicBlock(
          (conv1): Conv2d(64, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
          (downsample): Sequential(
            (0): Conv2d(64, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          )
        )
      )
      (conv1): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), b
ias=False)
      (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (maxpool2): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mod
e=False)
      (layer2): Sequential(
        (0): BasicBlock(
          (conv1): Conv2d(128, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
          (downsample): Sequential(
            (0): Conv2d(128, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          )
        )
        (1): BasicBlock(
```

```
          (conv1): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
      )
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), b
ias=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (maxpool3): MaxPool2d(kernel_size=2, stride=(2, 1), padding=(0, 1), dilation=
1, ceil_mode=False)
      (layer3): Sequential(
        (0): BasicBlock(
          (conv1): Conv2d(256, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
          (downsample): Sequential(
            (0): Conv2d(256, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          )
        )
        (1): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
        (2): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
```

```
        (3): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
        (4): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
      )
      (conv3): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), b
ias=False)
      (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (layer4): Sequential(
        (0): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
        (1): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
        (2): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
```

```
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
      )
      (conv4_1): Conv2d(512, 512, kernel_size=(2, 2), stride=(2, 1), padding=(0, 1),
bias=False)
      (bn4_1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
      (conv4_2): Conv2d(512, 512, kernel_size=(2, 2), stride=(1, 1), bias=False)
      (bn4_2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
    )
  )
  (AdaptiveAvgPool): AdaptiveAvgPool2d(output_size=(None, 1))
  (SequenceModeling): Sequential(
    (0): BidirectionalLSTM(
      (rnn): LSTM(512, 256, batch_first=True, bidirectional=True)
      (linear): Linear(in_features=512, out_features=256, bias=True)
    )
    (1): BidirectionalLSTM(
      (rnn): LSTM(256, 256, batch_first=True, bidirectional=True)
      (linear): Linear(in_features=512, out_features=256, bias=True)
    )
  )
  (Prediction): Linear(in_features=256, out_features=37, bias=True)
)
```

Step 7- Model Training:

Step 7.1- Model Training Stage 1- Freezing all CNN Parameters and training Bi-LSTM and Prediction Layer:

In the first stage of model training, **all parameters of the feature extraction layer (CNN) are frozen**. During this phase, only the weights of the Bi-LSTM and the prediction layer are updated. The initial layers of CNNs typically learn **general visual features** such as **edges, curves, and basic shapes**, which are largely transferable across datasets. Therefore, freezing the CNN helps preserve the useful representations learned during pretraining.

Training the entire model simultaneously from the beginning can lead to large weight updates that may overwrite these useful features and increase the risk of overfitting, particularly when the training dataset is small. By freezing the CNN layers initially, the model retains the previously learned feature representations while allowing the sequence modeling (Bi-LSTM) and prediction layers to adapt to the characteristics of the current dataset.

In the second stage of training, the entire model is unfrozen and trained end-to-end using a lower learning rate. This allows the network to fine-tune all parameters while minimizing the risk of overwriting the useful features learned during earlier training stages.

In [17]:
```
#Freezing all CNN layers for initial training:
for param in model.FeatureExtraction.parameters():
    param.requires_grad = False
```

In [18]:
```
#Checking total no of traininable and non trainable parameters:

total_params = sum(p.numel() for p in model.parameters())
trainable_params = sum(p.numel() for p in model.parameters() if p.requires_grad)

print("Total parameters:", total_params)
print("Trainable parameters:", trainable_params)
```

```
Total parameters: 47165701
Trainable parameters: 2901797
```

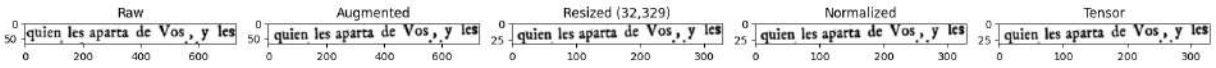
In [19]:
```
#Thus:
#~94% of the model is frozen (Paramters of TPS and CNN)
#~6% of the model is trainable (Parameters of Bi-LSTM and Predictor)

#Sanity Check- checking which layer's weights are trainable:
for name, param in model.named_parameters():
    if param.requires_grad:
        print(name)
```

```
SequenceModeling.0.rnn.weight_ih_l0
SequenceModeling.0.rnn.weight_hh_l0
SequenceModeling.0.rnn.bias_ih_l0
SequenceModeling.0.rnn.bias_hh_l0
SequenceModeling.0.rnn.weight_ih_l0_reverse
SequenceModeling.0.rnn.weight_hh_l0_reverse
SequenceModeling.0.rnn.bias_ih_l0_reverse
SequenceModeling.0.rnn.bias_hh_l0_reverse
SequenceModeling.0.linear.weight
SequenceModeling.0.linear.bias
SequenceModeling.1.rnn.weight_ih_l0
SequenceModeling.1.rnn.weight_hh_l0
SequenceModeling.1.rnn.bias_ih_l0
SequenceModeling.1.rnn.bias_hh_l0
SequenceModeling.1.rnn.weight_ih_l0_reverse
SequenceModeling.1.rnn.weight_hh_l0_reverse
SequenceModeling.1.rnn.bias_ih_l0_reverse
SequenceModeling.1.rnn.bias_hh_l0_reverse
SequenceModeling.1.linear.weight
SequenceModeling.1.linear.bias
Prediction.weight
Prediction.bias
```

In [20]:
```
#Visualizing preprocessing that is to be applied before modelling:
img_path, label = train_data[1]

img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)

plt.figure(figsize=(16,4))
plt.subplot(1,5,1)
plt.imshow(img, cmap="gray")
plt.title("Raw")
```

```
print("Raw:", img.shape, img.dtype, img.min(), img.max())

#Applying the augmentation pipeline: NOTE: Agmentation does not resize image.
aug = train_transform(image=img)["image"]
plt.subplot(1,5,2)
plt.imshow(aug, cmap="gray")
plt.title("Augmented")
print("Aug:", aug.shape, aug.dtype, aug.min(), aug.max())

#Resizing image:
imgH = 32 #as CRNN models expect fixed image height of 32 pixels
h, w = aug.shape
new_w = int(imgH * w / h) #rescaling width wrt height while preserving aspect ratio
resized = cv2.resize(aug, (new_w, imgH))

plt.subplot(1,5,3)
plt.imshow(resized, cmap="gray")
plt.title(f"Resized (32,{new_w})")
print("Resize:", resized.shape, resized.dtype, resized.min(), resized.max())


#Normalizing pixel values between 0-1:
norm = resized.astype(np.float32) / 255.0 #NOTE- neural networks require float as i
plt.subplot(1,5,4)
plt.imshow(norm, cmap="gray")
plt.title("Normalized")
print("Norm:", norm.shape, norm.dtype, norm.min(), norm.max())


#Converting image to tensor:
#Numpy array -> tensor (NN training in Pytorch req tensor inputs)
tensor_img = torch.from_numpy(norm).unsqueeze(0)
plt.subplot(1,5,5)
plt.imshow(tensor_img.squeeze().numpy(), cmap="gray")
plt.title("Tensor")
print("Tensor:", tensor_img.shape)

plt.tight_layout()
plt.show()
```

```
Raw: (71, 730) uint8 0 255
Aug: (71, 730) uint8 0 255
Resize: (32, 329) uint8 0 255
Norm: (32, 329) float32 0.0 1.0
Tensor: torch.Size([1, 32, 329])
```



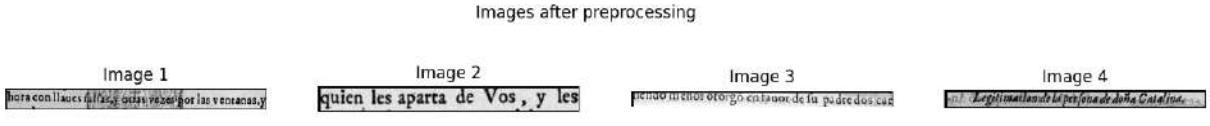
In [22]:
```
#Creating Dataset class:
class OCRDataset(Dataset):

    def __init__(self, data, transform=None, imgH=32):
        self.data = data
        self.transform = transform
        self.imgH = imgH
```

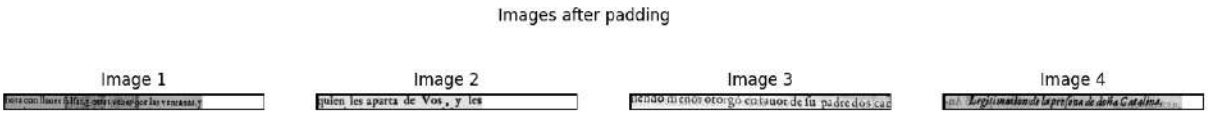
```
    def __len__(self):
        return len(self.data)

    def __getitem__(self, idx):

        img_path, label = self.data[idx]

        img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
        if img is None:
            raise ValueError(f"Failed to read image: {img_path}")

        if self.transform:
            img = self.transform(image=img)["image"]

        imgH = 32
        h, w = img.shape
        new_w = int(imgH * w / h)

        img = cv2.resize(img, (new_w, imgH))

        img = img.astype(np.float32) / 255.0

        tensor_img = torch.from_numpy(img).unsqueeze(0)

        return tensor_img, label
```

In [23]:
```
#Creating Dataset object from train data:
train_ds = OCRDataset(train_data, transform=train_transform)
```

In [24]:
```
#Next, applying padding to a few images and visualizing: (Padding is required as al
num_samples = 4  # how many images to visualize

processed_images = []
labels = []

#Applying preprocessing pipeline to replicate actual preprocessing before padding:
for i in range(num_samples):
    img_path, label = train_data[i]
    img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
    aug = train_transform(image=img)["image"]
    imgH = 32
    h, w = aug.shape
    new_w = int(imgH * w / h)
    resized = cv2.resize(aug, (new_w, imgH))
    norm = resized.astype(np.float32) / 255.0
    tensor_img = torch.from_numpy(norm).unsqueeze(0)

    processed_images.append(tensor_img)
    labels.append(label)


#Vizualizing preprocessed images for later comparison with padded images:
plt.figure(figsize=(16,2))
for i, img in enumerate(processed_images):
    ax = plt.subplot(1, num_samples, i+1)
```

```
    ax.imshow(img.squeeze().numpy(), cmap="gray")
    _, h, w = img.shape
    rect = patches.Rectangle(
        (0, 0), w, h,
        linewidth=2,
        edgecolor="black",
        facecolor="none")
    ax.add_patch(rect)
    plt.title(f"Image {i+1}")
    plt.axis("off")

plt.suptitle("Images after preprocessing")


#Applying padding:
widths = [img.shape[2] for img in processed_images] #list of all image widths for a
max_w = max(widths) #finding max image width

padded_images = []

for img in processed_images:
    c, h, w = img.shape
    padded = torch.ones((c, h, max_w)) #initializing empty white image- pixel value
                                       #step
    padded[:, :, :w] = img #inserting image to left side of the blank image created
    padded_images.append(padded)


# Vizualizing padded images:
plt.figure(figsize=(16,2))
for i, img in enumerate(padded_images):
    ax = plt.subplot(1, num_samples, i+1)
    ax.imshow(img.squeeze().numpy(), cmap="gray")
    _, h, w = img.shape
    rect = patches.Rectangle(
        (0, 0), w, h,
        linewidth=2,
        edgecolor="black",
        facecolor="none")
    ax.add_patch(rect)
    plt.title(f"Image {i+1}")
    plt.axis("off")

plt.suptitle("Images after padding")
plt.show()


print("Original widths:", widths)
print("Batch padded width:", max_w)
```






```
Original widths: [393, 329, 516, 473]
Batch padded width: 516
```

In [25]:
```python
#Writing function for padding:
def ocr_collate(batch):
    images, labels = zip(*batch)
    widths = [img.shape[2] for img in images]
    max_w = max(widths)
    padded_images = []
    for img in images:
        c, h, w = img.shape
        padded = torch.ones((c, h, max_w))
        padded[:, :, :w] = img
        padded_images.append(padded)
    images = torch.stack(padded_images) #converting padded images to a batch tensor
    return images, labels
```

In [26]:
```python
#For loading data and dividing data into batches for training:
train_loader = DataLoader(
    train_ds,
    batch_size=16,
    shuffle=True,
    num_workers=0,
    collate_fn=ocr_collate)
```

In [27]:
```python
#Removing the module. prefix from the key names in the downloaded model parameters
new_ckpt = {} #Dictionary for Storing Checkpoint values cleanly

for k, v in ckpt.items():
    new_key = k.replace("module.", "")
    new_ckpt[new_key] = v

#Next, loading the downloaded weights to the model architecture defined:
model.load_state_dict(new_ckpt, strict=False)

#The result shows that the downloaded model has extra keys related to the TPS layer
```

Out[27]:
```
_IncompatibleKeys(missing_keys=[], unexpected_keys=['Transformation.LocalizationNe
twork.conv.0.weight', 'Transformation.LocalizationNetwork.conv.1.weight', 'Transfo
rmation.LocalizationNetwork.conv.1.bias', 'Transformation.LocalizationNetwork.con
v.1.running_mean', 'Transformation.LocalizationNetwork.conv.1.running_var', 'Trans
formation.LocalizationNetwork.conv.1.num_batches_tracked', 'Transformation.Localiz
ationNetwork.conv.4.weight', 'Transformation.LocalizationNetwork.conv.5.weight',
'Transformation.LocalizationNetwork.conv.5.bias', 'Transformation.LocalizationNetw
ork.conv.5.running_mean', 'Transformation.LocalizationNetwork.conv.5.running_var',
'Transformation.LocalizationNetwork.conv.5.num_batches_tracked', 'Transformation.L
ocalizationNetwork.conv.8.weight', 'Transformation.LocalizationNetwork.conv.9.weig
ht', 'Transformation.LocalizationNetwork.conv.9.bias', 'Transformation.Localizatio
nNetwork.conv.9.running_mean', 'Transformation.LocalizationNetwork.conv.9.running_
var', 'Transformation.LocalizationNetwork.conv.9.num_batches_tracked', 'Transforma
tion.LocalizationNetwork.conv.12.weight', 'Transformation.LocalizationNetwork.con
v.13.weight', 'Transformation.LocalizationNetwork.conv.13.bias', 'Transformation.L
ocalizationNetwork.conv.13.running_mean', 'Transformation.LocalizationNetwork.con
v.13.running_var', 'Transformation.LocalizationNetwork.conv.13.num_batches_tracke
d', 'Transformation.LocalizationNetwork.localization_fc1.0.weight', 'Transformatio
n.LocalizationNetwork.localization_fc1.0.bias', 'Transformation.LocalizationNetwor
k.localization_fc2.weight', 'Transformation.LocalizationNetwork.localization_fc2.b
ias', 'Transformation.GridGenerator.inv_delta_C', 'Transformation.GridGenerator.P_
hat'])
```

In [28]:
```
#Printing out the weights corrosponding to the final prediction layer:
print(ckpt["module.Prediction.weight"])
print(ckpt["module.Prediction.bias"])
```

```
tensor([[-0.1551,  0.0523, -0.0088,  ..., -0.0074,  0.0192,  0.1050],
        [-0.6849,  0.0041, -0.0313,  ...,  0.0745, -0.0589,  0.1132],
        [-0.6352,  0.2659,  0.0186,  ..., -0.0424, -0.0151,  0.0443],
        ...,
        [-0.8234, -0.0581, -0.0670,  ...,  0.2254, -0.0645,  0.1287],
        [-0.4536,  0.1300,  0.0289,  ...,  0.0583,  0.0146,  0.2333],
        [-0.0908,  0.0457, -0.0886,  ..., -0.1506, -0.0219,  0.0957]])
tensor([ 0.1600, -1.9966, -1.6600, -2.1352, -2.0590, -2.3264, -2.3076, -2.3808,
        -2.2890, -2.4244, -2.1667, -0.2148, -0.8175, -0.6231, -0.5714, -0.1431,
        -0.8042, -0.7360, -0.4818, -0.2980, -1.6307, -1.1224, -0.4259, -0.6506,
        -0.2007, -0.3668, -0.7199, -1.8443, -0.2554, -0.2692, -0.1863, -0.5117,
        -0.9768, -0.8258, -1.4844, -0.8802, -1.7038])
```

In [29]:
```
#Printing the shape of the weights and bias in the final prediction layer:
print(ckpt["module.Prediction.weight"].shape)
print(ckpt["module.Prediction.bias"].shape)
```

```
torch.Size([37, 256])
torch.Size([37])
```

In [30]:
```
#Thus, the model predicts 37 characters,and the previous layer has 256 hidden units
#Replacing the final prediction layer as I want to predict the characters in the ch

#Computing no of characters in my character set:
num_classes = len(charset) + 1

print(num_classes)
```

```
79
```

In [31]:
```
#Replacing the final model layer:
import torch.nn as nn
model.Prediction = nn.Linear(
    model.Prediction.in_features, #Prediction.in_features=512 (size of input to pre
    num_classes # =79(size of output from prediction layer)
)
#The last layers weights will be randomly initialized
```

In [32]:
```
print(model) #As we can see, the output dimension of the prediction layer has been
```

```
Model(
  (FeatureExtraction): ResNet_FeatureExtractor(
    (ConvNet): ResNet(
      (conv0_1): Conv2d(1, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bi
as=False)
      (bn0_1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_s
tats=True)
      (conv0_2): Conv2d(32, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), b
ias=False)
      (bn0_2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_s
tats=True)
      (relu): ReLU(inplace=True)
      (maxpool1): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mod
e=False)
      (layer1): Sequential(
        (0): BasicBlock(
          (conv1): Conv2d(64, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
          (downsample): Sequential(
            (0): Conv2d(64, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          )
        )
      )
      (conv1): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), b
ias=False)
      (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (maxpool2): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mod
e=False)
      (layer2): Sequential(
        (0): BasicBlock(
          (conv1): Conv2d(128, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
          (downsample): Sequential(
            (0): Conv2d(128, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          )
        )
        (1): BasicBlock(
```

```
          (conv1): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
      )
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), b
ias=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (maxpool3): MaxPool2d(kernel_size=2, stride=(2, 1), padding=(0, 1), dilation=
1, ceil_mode=False)
      (layer3): Sequential(
        (0): BasicBlock(
          (conv1): Conv2d(256, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
          (downsample): Sequential(
            (0): Conv2d(256, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          )
        )
        (1): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
        (2): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
```

```
        (3): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
        (4): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
      )
      (conv3): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), b
ias=False)
      (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (layer4): Sequential(
        (0): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
        (1): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
        (2): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
```

```
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
          (relu): ReLU(inplace=True)
        )
      )
      (conv4_1): Conv2d(512, 512, kernel_size=(2, 2), stride=(2, 1), padding=(0, 1),
bias=False)
      (bn4_1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
      (conv4_2): Conv2d(512, 512, kernel_size=(2, 2), stride=(1, 1), bias=False)
      (bn4_2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
    )
  )
  (AdaptiveAvgPool): AdaptiveAvgPool2d(output_size=(None, 1))
  (SequenceModeling): Sequential(
    (0): BidirectionalLSTM(
      (rnn): LSTM(512, 256, batch_first=True, bidirectional=True)
      (linear): Linear(in_features=512, out_features=256, bias=True)
    )
    (1): BidirectionalLSTM(
      (rnn): LSTM(256, 256, batch_first=True, bidirectional=True)
      (linear): Linear(in_features=512, out_features=256, bias=True)
    )
  )
  (Prediction): Linear(in_features=256, out_features=79, bias=True)
)
```

In [33]:
```
#Defining CTC loss function for model training:
criterion = nn.CTCLoss(blank=0,
                       zero_infinity=True) # For numerical stability-To prevent los
                                           # and prediction- i.e, model output leng
```

In [34]:
```
#Building character encoder:
#Character encoder maps each character in character set to an index- the NN will le
#characters:
char_to_idx = {c: i+1 for i, c in enumerate(charset)}
idx_to_char = {i+1: c for i, c in enumerate(charset)}
```

In [35]:
```
#encoding function for labels- so that CTC loss can be applied:
def encode_labels(labels, char_to_idx):
    encoded = []
    lengths = []
    for label in labels:
        seq = [char_to_idx[c] for c in label if c in char_to_idx]
        encoded.extend(seq)
        lengths.append(len(seq)) #eg format- let labels=["cat", "dog", "hi"]
                                 # and after char_to_idx let seq no1=[3,1,20], seq n
                                 #Then targets = [3,1,20,4,15,7,8,9] and lengths = [
    return torch.LongTensor(encoded), torch.LongTensor(lengths) #list to tensor
```

In [36]:
```
#Applying model to one batch of training data to catch any Dimensionality issues:
```

```
#Getting batch of (images, labels) from train_loader:
images, labels = next(iter(train_loader))
```

In [45]:
```
#Printing shape of batch of 16 images and labels:
print(images.shape)
print(labels)
```

```
torch.Size([16, 1, 32, 1000])
('re apartar de mí, y después con la venida del amigo, el', 'E orden del Ilustre Señ
or Don Fran-', 'do la vida el mismo delinquente, queda toda la accion', 'Antonio, so
bre la fuerça y engaño, y quebrantamien-', 'dia', 'los bienes de doña Beatriz de Sil
ua, entre Atonio', 'parentibus. Sed vnusquisque pro peccato suo', 'DE', 'enseñar a l
os de nuestro siglo a ser estudiantes como lo han de ser, pues en el golfo de los ne
gocios,', 'rán al aire porque vos joya mía no queréis hacer ca-', 'los vocablos, com
o dezimos en Latin conteauno, y despicio por despreciar, pero mirando la propia', 'r
efirió al tiro que hizo Jacome, que luego que dieron las', 'cha correccion y tassa c
ierta que antes', 'tel, en que declara, que es mayor de 25. años, y que', 'por rendi
da, y yo como tal me echo', 'con el noticia a los Españoles de su propio lenguage, p
orque es impossible que se tenga')
```

In [38]:
```
#Passing one batch of images through CRNN model:

preds = model(images,None) #images= batch of images from data loader
#images is a tensor with shape: [batch_size, channels, height, width]
#Shape of output:(preds.shape) = [batch_size, sequence_length, num_classes]
#Preds is made of raw probailities: preds[i,j,k]- for ith image, on predicting jth
#set

preds.shape
```

Out[38]: torch.Size([16, 251, 79])

In [39]:
```
preds = preds.log_softmax(2) #Becomes matrix of Log probabilities of characters.
preds = preds.permute(1, 0, 2) #Rearrange to [seq_len, batch_size, num_classes] to

batch_size = preds.size(1) #Here, 16.
seq_len = preds.size(0) #Lenth of sequence the model predicts for each image

preds_size = torch.IntTensor([seq_len] * batch_size)

encoded, lengths = encode_labels(labels, char_to_idx) #Encoding labels according to
loss = criterion(preds, encoded, preds_size, lengths)
```

In [40]:
```
#Setting up optimizer:
optimizer = torch.optim.Adam(
    filter(lambda p: p.requires_grad, model.parameters()), #For Phase 1 of training
                                                           #- p.requires_grad=True.
    lr=0.001)
```

In [41]:
```
#Preparing validation dataset and data loader:
val_ds = OCRDataset(val_data, transform=None) #Transformation pipeline is not neede
val_loader = DataLoader(
    val_ds,
    batch_size=16,
    shuffle=False,   # no shuffling during validation
```

```
    num_workers=0,
    collate_fn=ocr_collate #the padding function used for train data as well.
)
```

In [42]:
```
#function to decode prediction-> text: I am using Greedy Decoding:

def greedy_decode(preds, idx_to_char, blank=0):

    # preds shape: [batch size, sequence length, number of character classes]
    preds = preds.detach().cpu() #Detach- removes tensor from gradient computation

    pred_indices = preds.argmax(2)   #Calculating which character(index) out of all
    #Pred_indices.shape=[batch size, sequence length]- matrix of predicted characte

    decoded_batch = [] #To store decoded strings

    for seq in pred_indices: #looping through 16(batch size) predicted image sequen
        prev = None #to remove duplicates
        decoded = "" #to hold the decoded image prediction

        for idx in seq: #looping through the predicted image sequence of a single i
            idx = idx.item() #Converting tensor to integer.
            if idx != blank and idx != prev: #if not blank and if not repetition:
                decoded += idx_to_char[idx]
            prev = idx
        decoded_batch.append(decoded)
    return decoded_batch
```

In [67]:
```
#Decoding the preds for this current batch:
decoded_preds = greedy_decode(preds, idx_to_char)
```

In [68]:
```
decoded_preds
# On re-running with the final predicted model (that I have fitted in the later ste
#matches the ground truth labels of this batch to a great extent.
```

Out[68]:
```
['DE',
 'quando marer este vilis persona vel suspecta de collu-',
 'varon, unico sucessor en el mayorazgo de su padre, mo-',
 'Colegio de San Saluador de Oviedo, el mayor de Salamanca, Sacristan',
 'cumentos, mereceran, quando hombres,',
 'confistitin veritate, & non infictione, & est socia di-',
 'de los señores Alcaldes, en que man-',
 'mucha',
 'mos en la trampa. BienSeque estas mis palabras se echa']
```

Step 7.2- Defining Model Evaluation Metrics::

Model Evaluation Metrics- CER and WER:

Character Error Rate (CER):

Measures how many characters in the predicted text are wrong compared to the ground truth. It is calculated using **Levenshtein distance**, which counts the minimum number of changes needed to convert one string into another. The allowed edits are:

1. Insertion (I) – adding an extra character
2. Deletion (D) – missing a character
3. Substitution (S) – replacing one character with another

Formula:

$$CER = \frac{S + D + I}{N}$$

Where:

- S = Number of substitutions
- D = Number of deletions
- I = Number of insertions
- N = Number of characters in the ground truth

Example use: Ground truth="cat", Prediction="mat", then to go from GT to Prediction, we need 1 substitution- "m" to "c". N = 3 (length of GT)

Then, CER = 0.33333

Word Error Rate (WER):

Measures how many words in the predicted text are wrong compared to the ground truth. Uses the same formuala as CER, just words are the units of measurement as opposed to individual characters.

Example use: Ground truth="cat is sleepy", Prediction="cat is slepy", then to go from GT to Prediction, we need 1 substitution- "slepy" to "sleepy". N=3 (length of GT)

Then, WER = 0.33333

Why are we evaluating model using CER/WER and not the CTC loss value?

The model is trained by minimizing the CTC loss-i.e, the parameters are updated using this loss in the training process. A lower CTC loss also means that the model assigns higher probability to the correct sequence.

However, this probability is calculated before decoding and includes many possible alignments.CTC uses probabilities of all valid alignments, not just the final decoded text. The final output also depends on the decoding method used, not only the predicted alignments.

CER takes into account only the final decoded text- thus evaluating performance of the model+decoder.This metric is important because, in OCR systems, what ultimately matters is the accuracy of the final decoded prediction.

Thus, in the model training, at each step, the model which gives lowest CER on the Validation dataset(and not the model with the lowest Validation loss) is picked up up for the subsequent steps.

In [46]:
```
#Defining Levenshtein distance function for CER calculation:
def levenshtein(a, b): #Inputs-(Predicted text, GT text)
    m = len(a)
    n = len(b)
    dp = [[0]*(n+1) for _ in range(m+1)] #creating 2D matrix of zeroes of dimension
    #dp[i][j] = minimum edit distance between a[:i] and b[:j] or how many edits are
    #first j characters of b

    #[0,0] position corrosponds to blanks.

    #Filling first column and row:
    for i in range(m+1):
        dp[i][0] = i  #no of deletions needed to convert the first i characters of
    for j in range(n+1):
        dp[0][j] = j  #no of insertions neede to convert an empty string into the f

    for i in range(1, m+1):
        for j in range(1, n+1):
            cost = 0 if a[i-1] == b[j-1] else 1
            dp[i][j] = min(
                dp[i-1][j] + 1,      # deletion
                dp[i][j-1] + 1,      # insertion
                dp[i-1][j-1] + cost  # substitution
            )
    return dp[m][n] #Result= Bottom-right of matrix
```

In [47]:
```
#Defining Character Error Rate calculation function for a batch:
def calculate_cer(pred_texts, gt_texts):
    total_dist = 0 #to store total edit distance for batch
    total_chars = 0 # to store total number of characters in all ground truth text.
    for pred, gt in zip(pred_texts, gt_texts):
        dist = levenshtein(pred, gt)
        total_dist += dist
        total_chars += len(gt)
    if total_chars == 0:
        return 0
    return total_dist / total_chars
```

In [48]:
```
#Function to calculate Word Error Rate:- Same logic as CER- applied to words in pla
def calculate_wer(pred_texts, gt_texts):
    total_dist = 0
    total_words = 0
    for pred, gt in zip(pred_texts, gt_texts):
        pred_words = pred.split()
        gt_words = gt.split()
```

```
        dist = levenshtein(pred_words, gt_words)
        total_dist += dist
        total_words += len(gt_words)
    if total_words == 0:
        return 0
    return total_dist / total_words
```

In [49]:

```
#Stage 1 Model Training:(Training with CNN frozen, for 10 Epochs)
num_epochs = 20

#initiating variables to store the lowest total loss/CER accross all epochs:
best_train_loss = float("inf")
best_val_loss = float("inf")
best_cer = float("inf")

for epoch in range(num_epochs):

    #Training Phase:
    model.train()
    total_loss = 0

    for images, labels in tqdm(train_loader, desc=f"Epoch {epoch+1}/{num_epochs}"):
        preds = model(images, None)  #Passing a batch of images to model
        log_preds = preds.log_softmax(2) #Character probabilities calculated
        ctc_preds = log_preds.permute(1,0,2)  #Permutation for compatibility with e
        seq_len = ctc_preds.size(0)
        batch_size = ctc_preds.size(1)
        preds_size = torch.IntTensor([seq_len] * batch_size)
        encoded, lengths = encode_labels(labels, char_to_idx) #Converting ground tr
        loss = criterion(ctc_preds, encoded, preds_size, lengths)

        #Updating model parameters:
        optimizer.zero_grad() #clearing previously stored gradients from optimizer
        loss.backward() #Carries out backpropagation i.e, calculates gradients- del
        optimizer.step() #Updating parameters using calculated gradients-new_para =
        total_loss += loss.item() #Total loss for current epoch

    train_loss = total_loss / len(train_loader) #Average training loss per batch fo

    #Validation phase:
    model.eval()
    val_loss = 0 #Variable to store total validation loss in current epoch
    all_preds = [] #list to store all predicted texts
    all_labels = [] #list to store all ground truth labels

    with torch.no_grad(): #Gradients not computed
        for images, labels in tqdm(val_loader, desc="Validation"):

            #Same as training loop:

            preds = model(images, None)
            log_preds = preds.log_softmax(2)
            ctc_preds = log_preds.permute(1,0,2)
            seq_len = ctc_preds.size(0)
            batch_size = ctc_preds.size(1)
            preds_size = torch.IntTensor([seq_len] * batch_size)
```

```
            encoded, lengths = encode_labels(labels, char_to_idx)
            loss = criterion(ctc_preds, encoded, preds_size, lengths)
            val_loss += loss.item()

            #For validation data, we want to calculate CER:
            decoded_preds = greedy_decode(preds, idx_to_char) #Greedy decoding pred
            all_preds.extend(decoded_preds) #Saving predicted values
            all_labels.extend(labels)

    val_loss = val_loss / len(val_loader) #Average validation loss per batch in cur

    cer = calculate_cer(all_preds, all_labels) #Calculating Validation CER for epoc

    #Saving best models for later use:
    if train_loss < best_train_loss:
        best_train_loss = train_loss
        torch.save(model.state_dict(), "greedy_best_train_loss_model.pth")
        print("Saved best TRAIN loss model (greedy)")

    if val_loss < best_val_loss:
        best_val_loss = val_loss
        torch.save(model.state_dict(), "greedy_best_val_loss_model.pth")
        print("Saved best VALIDATION loss model (greedy)")

    if cer < best_cer:
        best_cer = cer
        torch.save(model.state_dict(), "greedy_best_cer_model.pth")
        print("Saved best CER model (greedy)")

    # Printing out Loss and CER values for each Epoch:
    print(f"\nEpoch {epoch+1}/{num_epochs}")
    print(f"Train Loss : {train_loss:.4f}")
    print(f"Val Loss   : {val_loss:.4f}")
    print(f"CER        : {cer:.4f}")

    #Printing out few predictions every epoch from Validation Set (Sanity Check):
    print("\nSample Predictions:\n")
    pairs = list(zip(all_preds, all_labels))
    sample_pairs = random.sample(pairs,min(5, len(pairs)))
    for pred, gt in sample_pairs:
        print("GT   :", gt)
        print("Pred :", pred)
        print()
```

```
alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:13<00:00,  4.38s/it]
```

```
Saved best TRAIN loss model (greedy)
Saved best VALIDATION loss model (greedy)
Saved best CER model (greedy)

Epoch 1/20
Train Loss : 4.6648
Val Loss   : 2.5087
CER        : 0.4759

Sample Predictions:

GT   : tonio ha de ser condenado a q avien
Pred : coniohadelercondenadoagaic-

GT   : C
Pred : last

GT   : MADRE, que se porten como sus
Pred : adre eportencomo -

GT   : parte para el juez que lo sentenciare, y la
Pred : parteparaeliieeuelolentenciarel-

GT   : daron deuiendo, quando murio doña Beatriz
Pred : daron deirdosandomriodonaeatri-

alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.25s/it]
Saved best TRAIN loss model (greedy)
Saved best VALIDATION loss model (greedy)
Saved best CER model (greedy)

Epoch 2/20
Train Loss : 1.4825
Val Loss   : 1.5025
CER        : 0.2601

Sample Predictions:

GT   : conclusa la causa
Pred : conelusa la causa

GT   : Matt. I8. proponeis por norma del can-
Pred : matfa ro poneis por nor ma de s can

GT   : to del lenguaje Latino, engaño en los venideros a muchos: y assi el bienavent
urado San Isidoro, que
Pred : de ngi, tno engaho enios reniueros ams sestya sicibisnasirados ldoruq

GT   : no sea heredera de su marido, se prefiere en la acusacion
Pred : lo ealieradarade liamaride e prefiercienlaaeusai

GT   : Santiago.
Pred : Santiago
```

```
alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.19s/it]
Saved best TRAIN loss model (greedy)
Saved best VALIDATION loss model (greedy)
Saved best CER model (greedy)

Epoch 3/20
Train Loss : 0.9814
Val Loss   : 1.1454
CER        : 0.2448

Sample Predictions:

GT   : MADRE, que se porten como sus
Pred : Madre, que se porten como sus

GT   : Y de la pena que se deve imponer por los dichos de-
Pred : Yde lapena que se deueimponer porlos dichos de

GT   : dre de Iacinto Gomez, para que diesse bienes de su
Pred : dre de lacinto Gomez para que diessebienes de su

GT   : testimonio de la particion, que se auia hecho de
Pred : tetimniode la parrieion, gestaiheehode

GT   : parte para el juez que lo sentenciare, y la
Pred : parte paraeljuez que lo sentenciare, y l

alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.26s/it]
```

```
Saved best TRAIN loss model (greedy)
Saved best VALIDATION loss model (greedy)
Saved best CER model (greedy)

Epoch 4/20
Train Loss : 0.8400
Val Loss   : 1.0168
CER        : 0.2006

Sample Predictions:

GT   : C
Pred : s

GT   : fee en publica forma como por corrector
Pred : feen publica forma como por corrector

GT   : tido; y parecidole mal, pues la acepcion del vocablo en su propia Etymologia
es la verdadera signi-
Pred : ado, precidose a a pes laa ce t ion del vocablo en  propia s,noloai es la tra

GT   : Iulius Clar. q. 58. num. 31. vers. contingit etiam. Bona-
Pred : tuliur e lae qs , numl3 t ves s contingic etiam, Bona

GT   : rradas las puertas con llaves, y usado de otras falsas, y
Pred : radas las puestas con llaues, y v lado de otras falsas, ,

alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:14<00:00,  4.71s/it]
```

```
Saved best TRAIN loss model (greedy)
Saved best VALIDATION loss model (greedy)
Saved best CER model (greedy)

Epoch 5/20
Train Loss : 0.7431
Val Loss   : 0.7465
CER        : 0.1804

Sample Predictions:

GT   : tido; y parecidole mal, pues la acepcion del vocablo en su propia Etymologia es la verdadera signi-
Pred : sdo,  precidosenaa pes laacetion del vocablo en  propia stynologiaes lavtri

GT   : y que se pidio su aprouacion, y que se auia dado
Pred : y que se pidio suaptovacion, y queseavia dado

GT   : han escrito libros Criticos en nuestros tiempos, han procurado adornarlos con verdaderas, y doctas
Pred : aane scrito libros Criticos en noestros tiempos, sno peeinado aderoaes cen ve edaeaa , c at s

GT   : Y de la pena que se deve imponer por los dichos de-
Pred : Yde la pena que se deueimponer por los dichos de

GT   : cion, y Capellan de su Magestad, y no he allado en el cosa contraria a
Pred : cion, y capelslan de s Magestad, y nohe hallado en el cose, contraria a

alidation: 100%|████████████████████████████████████████████████████████| 3/3 [00:13<00:00,  4.61s/it]
```

```
Saved best TRAIN loss model (greedy)
Saved best CER model (greedy)

Epoch 6/20
Train Loss : 0.6348
Val Loss   : 0.7576
CER        : 0.1770

Sample Predictions:

GT   : no sea heredera de su marido, se prefiere en la acusacion
Pred : ho sea licraderade siamaride e prefiercen laacusain

GT   : migo. Todo os sirva por bien, y aquí con desentrañable
Pred : migo Tado p, srua por hieny aqui con desintranable

GT   : los Corregidores, Assistente, Governado-
Pred : los Corregidores, Asistente, Gouernado

GT   : muerte de su marido, e el mari
Pred : muerte de sumarido, eel mar

GT   : Escriuano, en 19. de Enero de 1640. por Iacinto
Pred : Eseruanoen  de Encro d, placint,

alidation: 100%|████████████████████████████████████████████████████████
████| 3/3 [00:14<00:00,  4.67s/it]
Saved best TRAIN loss model (greedy)
Saved best VALIDATION loss model (greedy)

Epoch 7/20
Train Loss : 0.5991
Val Loss   : 0.6666
CER        : 0.1863

Sample Predictions:

GT   : mandamiento para cobrar las penas pecuniarias, y
Pred : mandamiento para cobrar las penas pecuniarias, y

GT   : los Corregidores, Assistente, Governado-
Pred : los Corregidores, Asistente, Governado

GT   : parte que le toca de la dote de su madre, y tambien
Pred : parte que le toca de la dote de su madre, y tambie

GT   : parte para el juez que lo sentenciare, y la
Pred : parte para el juiez que lo sentenciare, y la

GT   : han escrito libros Criticos en nuestros tiempos, han procurado adornarlos con
verdaderas, y doctas
Pred : aayestrito libros Criticos en noestros ciempos, soo peeiado adereaos cen yeed
aa,y c tt s
```

```
alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:14<00:00,  4.73s/it]
Saved best TRAIN loss model (greedy)
Saved best VALIDATION loss model (greedy)
Saved best CER model (greedy)

Epoch 8/20
Train Loss : 0.5455
Val Loss   : 0.6247
CER        : 0.1672

Sample Predictions:

GT   : dre de Iacinto Gomez, para que diesse bienes de su
Pred : dre de lacinto Gomez para que diesse bienes de su

GT   : fee en publica forma como por corrector
Pred : fee en publica forma como por corrector

GT   : primer lugar en acusar, le tiene priuative en perdonar,
Pred : primer lugar en acusar,le tiene priuatiue en perdonara

GT   : to del lenguaje Latino, engaño en los venideros a muchos: y assi el bienavent
urado San Isidoro, que
Pred : o dessengua3 aaio, engaño en los eniueros amssos,y asicibisnasisrados ldoruq

GT   : crianza de la niñez. Assi sea,
Pred : crianza de la Niñez, Asi sea,

alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:13<00:00,  4.54s/it]
```

```
Saved best TRAIN loss model (greedy)
Saved best CER model (greedy)

Epoch 9/20
Train Loss : 0.5117
Val Loss   : 0.6852
CER        : 0.1549

Sample Predictions:

GT   : los Corregidores, Assistente, Governado-
Pred : los Corregidores, Asistente, Governado

GT   : to del lenguaje Latino, engaño en los venideros a muchos: y assi el bienavent
urado San Isidoro, que
Pred : to dessengna3e aatio, engaño en los veniueros amissos,yasicibisnausiscirado s
n ldoruqe

GT   : no sea heredera de su marido, se prefiere en la acusacion
Pred : ho sea slieradernde sia maride, se prefiercen la acusacin-

GT   : testimonio de la particion, que se auia hecho de
Pred : testimonio de la parricion, quese a i, hecho de

GT   : mandamiento para cobrar las penas pecuniarias, y
Pred : mandamiento para cobrar las penas pecuniarias, y


alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:13<00:00,  4.56s/it]
```

```
Saved best TRAIN loss model (greedy)

Epoch 10/20
Train Loss : 0.4949
Val Loss   : 0.6480
CER        : 0.1657

Sample Predictions:

GT   : Y de la pena que se deve imponer por los dichos de-
Pred : Y de la pena que se deueimponer por los dichos de

GT   : gunas Etymologias que quiso deduzir de los vocablos que el fingio ser origina
riamente Latinos, no
Pred : gunsa, ur nalogas que quisto de uazir de los vocablos sge lgangio ser ongiam
iagmnaa lea tinosno

GT   : Escriuano, en 19. de Enero de 1640. por Iacinto
Pred : Eseriuano en  de Ecro6 par sacinte

GT   : ca de Caycedo su hija, donzella noble y honesta, a
Pred : ta de Ccaycedo su hija, donzella noble y hon esta, a

GT   : han escrito libros Criticos en nuestros tiempos, han procurado adornarlos con
verdaderas, y doctas
Pred : aanv estrito libros Criticos en noestros tiempos, snn pat e eiunado a deriear
les cen ycr da aa aa, y c tt s


alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:19<00:00,  6.45s/it]
```

```
Saved best TRAIN loss model (greedy)
Saved best CER model (greedy)

Epoch 11/20
Train Loss : 0.4747
Val Loss   : 0.7516
CER        : 0.1445

Sample Predictions:

GT   : parece de sus palabras, que traducidas en castellano so-
Pred : pareee de us palabras, que traducidas en Castellare so.

GT   : oficio de la mesma manera: como el cuchillo; y las tixeras, se hizieron ambos para cortar, pero no
Pred : sicio de la me sena manoa, eomn elcichillo ; y las ciceas sieet izieren ambo, para certar yero no

GT   : C
Pred : . C

GT   : to del lenguaje Latino, engaño en los venideros a muchos: y assi el bienavent urado San Isidoro, que
Pred : to desengua3e aatio, engano en los venideros amlos,yasicibienauciatirado s  l udorque

GT   : ca de Caycedo su hija, donzella noble y honesta, a
Pred : ta de caycedo suhija, donzella noble y hon estas a

alidation: 100%|████████████████████████████████████████████████████████████| 3/3 [00:19<00:00,  6.44s/it]
```

```
Saved best TRAIN loss model (greedy)
Saved best CER model (greedy)

Epoch 12/20
Train Loss : 0.4486
Val Loss   : 0.7613
CER        : 0.1352

Sample Predictions:

GT   : villete al
Pred : villete al.

GT   : tonio ha de ser condenado a q avien
Pred : conio ha de ser condenado a q avien

GT   : crianza de la niñez. Assi sea,
Pred : crianza de la Niñez. Asi sea-,

GT   : gunas Etymologias que quiso deduzir de los vocablos que el fingio ser origina
riamente Latinos, no
Pred : gus a r nologas que quiso deuzir de los vocablos qbee langio lerongin iagene
leosinos no

GT   : el disseño, la luz, y el exemplo,
Pred : el disseño, Ia luz, y el exemplo,

alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:11<00:00,  3.83s/it]
Saved best TRAIN loss model (greedy)

Epoch 13/20
Train Loss : 0.4033
Val Loss   : 0.9204
CER        : 0.1411

Sample Predictions:

GT   : to del lenguaje Latino, engaño en los venideros a muchos: y assi el bienavent
urado San Isidoro, que
Pred : to dessangua3e aatino, engaño en los veniueros a massos,y asi ci bienauaiascr
ado l ludoruaque

GT   : parte para el juez que lo sentenciare, y la
Pred : parte para el juiez que lo sentenciare, y la-

GT   : daron deuiendo, quando murio doña Beatriz
Pred : daron devierdo, quando murio doña Beatriz

GT   : rradas las puertas con llaves, y usado de otras falsas, y
Pred : tradas las puestas con llaves, y vlado de otras falsas, y

GT   : mandamiento para cobrar las penas pecuniarias, y
Pred : mandamiento para cobrar las penas pecuniarias, y
```

```
alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.15s/it]
Saved best TRAIN loss model (greedy)
Saved best VALIDATION loss model (greedy)

Epoch 14/20
Train Loss : 0.3801
Val Loss   : 0.5788
CER        : 0.1362

Sample Predictions:

GT   : oficio de la mesma manera: como el cuchillo; y las tixeras, se hizieron ambos
para cortar, pero no
Pred : sicin de la mesena manoa, esmn el cichislo, y y las cixeaas, see izieren amb
o, para certar, yero no

GT   : cion, y Capellan de su Magestad, y no he allado en el cosa contraria a
Pred : cion, y Capean de su Magestad, y no he hallado en el cose, contraria a

GT   : parte para el juez que lo sentenciare, y la
Pred : parte para el juiez que lo sentenciare, y l-

GT   : migo. Todo os sirva por bien, y aquí con desentrañable
Pred : migo Tado sirua por hieny Aqui con desintrañable

GT   : momento su remission, vt fere in terminis resoluune
Pred : momentoisu remissión, vefere in terminis resolunc

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Saved best TRAIN loss model (greedy)
Saved best VALIDATION loss model (greedy)
Saved best CER model (greedy)

Epoch 15/20
Train Loss : 0.3707
Val Loss   : 0.5563
CER        : 0.1342

Sample Predictions:

GT   : Santiago.
Pred : Santiago

GT   : tonio ha de ser condenado a q avien
Pred : conio ha de ser condenado a q avien

GT   : momento su remission, vt fere in terminis resoluune
Pred : momentoisu remissióna vefere in terminis resolunc

GT   : el disseño, la luz, y el exemplo,
Pred : el disseño, la luz, y el ejemplo,

GT   : villete al
Pred : villete al.

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Saved best TRAIN loss model (greedy)
Saved best VALIDATION loss model (greedy)

Epoch 16/20
Train Loss : 0.3632
Val Loss   : 0.5354
CER        : 0.1347

Sample Predictions:

GT   : oficio de la mesma manera: como el cuchillo; y las tixeras, se hizieron ambos
para cortar, pero no
Pred : sficin de la mesena ma no ea, eoman el crichislo, y y las ciyeas y Ee izieren
ambo, para certar, y ero no

GT   : 12 Otra carta de pago, otorgada en esta Vi-
Pred :  12 Otra carta de pago, otorgada en esta Vi-

GT   : el disseño, la luz, y el exemplo,
Pred : el disseño, Ia luz, y el exemplo,

GT   : Y de la pena que se deve imponer por los dichos de-
Pred : Y de la pena que se deuc imponer por los dichos de

GT   : Matt. I8. proponeis por norma del can-
Pred : Matr, 18. proponeis por nor ma del can
```

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Saved best TRAIN loss model (greedy)
Saved best VALIDATION loss model (greedy)
Saved best CER model (greedy)

Epoch 17/20
Train Loss : 0.3618
Val Loss   : 0.5003
CER   : 0.1332

Sample Predictions:

GT   : MADRE, que se porten como sus
Pred : MADrE, que se porten como sus

GT   : migo. Todo os sirva por bien, y aquí con desentrañable
Pred : migo Todo p serua porhiemy Aqui con desentrañable

GT   : C
Pred : G

GT   : los que han deseado mucho cumplir con las obligaciones de aquel oficio, de cu
yas obras no ten-
Pred : los que ñar deseado mucho cumplir con las obligaciones de aque loricigure cuy
as obras no ten

GT   : primer lugar en acusar, le tiene priuative en perdonar,
Pred : primer lugar en acusar, le tiene priuatiue en perdona raI

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Saved best TRAIN loss model (greedy)
Saved best CER model (greedy)

Epoch 18/20
Train Loss : 0.3218
Val Loss   : 0.6908
CER        : 0.1244

Sample Predictions:

GT   : primer lugar en acusar, le tiene priuative en perdonar,
Pred : primer lugar en acusar, le tiene priunatiue en perdonaraI

GT   : la cosa mas recebida que ay en los
Pred : la cosa mas recebida que ay en los

GT   : no sea heredera de su marido, se prefiere en la acusacion
Pred : no sea heraderade su marido, le prefiercen la acusacin

GT   : fee en publica forma como por corrector
Pred : feeen publica forma como por corrector

GT   : alomenos concurrir con ella: y que de aqui infirio, que pues
Pred : alomenos concurtrir co ellay de acue infirió, que pues


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Epoch 19/20
Train Loss : 0.3230
Val Loss   : 0.5062
CER        : 0.1288

Sample Predictions:

GT   : daron deuiendo, quando murio doña Beatriz
Pred : daron deuicido, quando murio doña Beatriz

GT   : Y de la pena que se deve imponer por los dichos de-
Pred : Y de la pena que se deue imponer por los dichos de

GT   : dre de Iacinto Gomez, para que diesse bienes de su
Pred : dre de Iacinto Gomez para que diesse bienes de su

GT   : Iulius Clar. q. 58. num. 31. vers. contingit etiam. Bona-
Pred : Tuliu, o lat, q. 3&, num. 13. vei ficontingic etiam. Bona3

GT   : el disseño, la luz, y el exemplo,
Pred : el disseño, la luz, y el exemplo,


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Saved best TRAIN loss model (greedy)

Epoch 20/20
Train Loss : 0.3018
Val Loss   : 0.5379
CER        : 0.1264

Sample Predictions:

GT   : la cosa mas recebida que ay en los
Pred : la cosa mas recebida que ay en los

GT   : oficio de la mesma manera: como el cuchillo; y las tixeras, se hizieron ambos
para cortar, pero no
Pred : sicin de la mesena manoa, eomn el cichislo;y y las ciydar Eeizieren ambo, par
a certar, yero no

GT   : crianza de la niñez. Assi sea,
Pred : crianza de la Niñez Asi sea-

GT   : parte que le toca de la dote de su madre, y tambien
Pred : parte que le toca de la dote de su madre, y tambi-

GT   : villete al
Pred : villete al
```

Stage 1 Training (CNN Weights Frozen)- Final Training Results:

The model was trained for 20 Epochs and the **Character Error Rate** decreased from 47% in Epoch 1 to **12.44% in Epoch 18**.

Thus, after training, about 12% of characters in the predicted text differ from the ground truth. The Character Error Rate (CER) decreased sharply during the initial epochs, followed by a gradual plateau.

In [50]:
```python
#Loading best model:
model.load_state_dict(torch.load("greedy_best_cer_model.pth"))
```

Out[50]:
```
<All keys matched successfully>
```

Step 7.3- Model Training Stage 2 (Training Full Model):

Unfreezing the CNN layers and training the complete model with a lower learning rate:

In [52]:
```python
#Unfreezing the CNN to train whole model:
for param in model.FeatureExtraction.parameters():
    param.requires_grad = True
```

In [53]:
```python
#Reducing the learning rate to prevent overwriting useful weights:
optimizer = torch.optim.Adam([
    {"params": model.FeatureExtraction.parameters(), "lr": 1e-5},
```

```
    {"params": model.SequenceModeling.parameters(), "lr": 1e-4},
    {"params": model.Prediction.parameters(), "lr": 1e-4}
])
```

In [54]:

```
#Stage 2 Model Training:25 Epochs
#Same loop as before

num_epochs = 25

best_train_loss = float("inf")
best_val_loss = float("inf")
best_cer = float("inf")

for epoch in range(num_epochs):
    #Training:
    model.train()
    total_loss = 0

    for images, labels in tqdm(train_loader, desc=f"Epoch {epoch+1}/{num_epochs}"):

        preds = model(images, None)
        log_preds = preds.log_softmax(2)
        ctc_preds = log_preds.permute(1,0,2)
        seq_len = ctc_preds.size(0)
        batch_size = ctc_preds.size(1)
        preds_size = torch.IntTensor([seq_len] * batch_size)
        encoded, lengths = encode_labels(labels, char_to_idx)
        loss = criterion(ctc_preds, encoded, preds_size, lengths)

        optimizer.zero_grad()
        loss.backward()
        optimizer.step()
        total_loss += loss.item()
    train_loss = total_loss / len(train_loader)

    #Validation:
    model.eval()

    val_loss = 0
    all_preds = []
    all_labels = []

    with torch.no_grad():
        for images, labels in tqdm(val_loader, desc="Validation"):
            preds = model(images, None)
            log_preds = preds.log_softmax(2)
            ctc_preds = log_preds.permute(1,0,2)
            seq_len = ctc_preds.size(0)
            batch_size = ctc_preds.size(1)
            preds_size = torch.IntTensor([seq_len] * batch_size)
            encoded, lengths = encode_labels(labels, char_to_idx)
            loss = criterion(ctc_preds, encoded, preds_size, lengths)
            val_loss += loss.item()

            decoded_preds = greedy_decode(preds, idx_to_char)
```

```
            all_preds.extend(decoded_preds)
            all_labels.extend(labels)

    val_loss = val_loss / len(val_loader)

    cer = calculate_cer(all_preds, all_labels)

    if train_loss < best_train_loss:
        best_train_loss = train_loss
        torch.save(model.state_dict(), "stage2_best_train_loss_model.pth")
        print("Saved best TRAIN loss model (stage2)")

    if val_loss < best_val_loss:
        best_val_loss = val_loss
        torch.save(model.state_dict(), "stage2_best_val_loss_model.pth")
        print("Saved best VALIDATION loss model (stage2)")

    if cer < best_cer:
        best_cer = cer
        torch.save(model.state_dict(), "stage2_best_cer_model.pth")
        print("Saved best CER model (stage2)")

    print(f"\nEpoch {epoch+1}/{num_epochs}")
    print(f"Train Loss : {train_loss:.4f}")
    print(f"Val Loss   : {val_loss:.4f}")
    print(f"CER        : {cer:.4f}")


    print("\nSample Predictions:\n")
    pairs = list(zip(all_preds, all_labels))
    sample_pairs = random.sample(pairs,min(5, len(pairs)))

    for pred, gt in sample_pairs:
        print("GT   :", gt)
        print("Pred :", pred)
        print()
```

```
alidation: 100%|████████████████████████████████████████████████████████████
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Saved best TRAIN loss model (stage2)
Saved best VALIDATION loss model (stage2)
Saved best CER model (stage2)

Epoch 1/25
Train Loss : 0.2727
Val Loss   : 0.5849
CER        : 0.0988

Sample Predictions:

GT   : parte que le toca de la dote de su madre, y tambien
Pred : parte que le toca de la dote de su madre, y tambi-

GT   : cion, y Capellan de su Magestad, y no he allado en el cosa contraria a
Pred : cion, y Capelan desu Magestad, y no he hallado en el cosa contraria a

GT   : Matt. I8. proponeis por norma del can-
Pred : Matr, 18. proponeis por norma del can-

GT   : de la casa, y llevandola a otras partes para usar mal de
Pred : de la casa, y llcvando la ajotras partes para vlar malde

GT   : MADRE, que se porten como sus
Pred : MADRE, que se porten como sus

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Saved best TRAIN loss model (stage2)
Saved best CER model (stage2)

Epoch 2/25
Train Loss : 0.2519
Val Loss   : 0.7348
CER        : 0.0954

Sample Predictions:

GT   : mandamiento para cobrar las penas pecuniarias, y
Pred : mandamiento para cobrar las penas pecuniarias, y

GT   : C
Pred : l  GG

GT   : momento su remission, vt fere in terminis resoluune
Pred : momento su remission, vefere in terminis resoluunc

GT   : de la casa, y llevandola a otras partes para usar mal de
Pred : de la casa, y llcvando la ajotras partes para vlar malde

GT   : gunas Etymologias que quiso deduzir de los vocablos que el fingio ser origina
riamente Latinos, no
Pred : gunas Cernalogias que quiso de uuzir de los vocablos que essungioser originar
iamente Latinos no
```

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Saved best TRAIN loss model (stage2)
Saved best VALIDATION loss model (stage2)
Saved best CER model (stage2)

Epoch 3/25
Train Loss : 0.2264
Val Loss   : 0.4977
CER   : 0.0885

Sample Predictions:

GT   : los que han deseado mucho cumplir con las obligaciones de aquel oficio, de cu
yas obras no ten-
Pred : los que han deseado mucho cumplir con las obligaciones de aque l oñicio, ue c
uyas obras no ten

GT   : daron deuiendo, quando murio doña Beatriz
Pred : daron deuiendo, quando murio doña Beatriz

GT   : primer lugar en acusar, le tiene priuative en perdonar,
Pred : primer lugar en acusar, le tiene priuatiue en perdonara

GT   : parte para el juez que lo sentenciare, y la
Pred : parte para el juez que lo sentenciare, y la

GT   : to del lenguaje Latino, engaño en los venideros a muchos: y assi el bienavent
urado San Isidoro, que
Pred : to dessengua3e Latino, engaho en los venideros a mucsos yasi ci bienauc initu
irado s n lsdoro, que

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Saved best TRAIN loss model (stage2)
Saved best VALIDATION loss model (stage2)
Saved best CER model (stage2)

Epoch 4/25
Train Loss : 0.2134
Val Loss   : 0.4473
CER        : 0.0846

Sample Predictions:

GT   : parece de sus palabras, que traducidas en castellano so-
Pred : pareee desus palabras, que traducidas en Castellano so-

GT   : no sea heredera de su marido, se prefiere en la acusacion
Pred : no sea heroderade su marido, le prefiercen la acusació

GT   : to del lenguaje Latino, engaño en los venideros a muchos: y assi el bienavent
urado San Isidoro, que
Pred : to dessngua3e Latino, engaho en los venideros a muchos yasici bienaucinituira
do s n lsidoru,que

GT   : y que se pidio su aprouacion, y que se auia dado
Pred : y que se pidio su aprouacion, y que se auia dado

GT   : momento su remission, vt fere in terminis resoluune
Pred : momento su remission, ve fere in terminis resoluunc


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Saved best TRAIN loss model (stage2)
Saved best CER model (stage2)

Epoch 5/25
Train Loss : 0.2062
Val Loss   : 0.4641
CER        : 0.0806

Sample Predictions:

GT   : no sea heredera de su marido, se prefiere en la acusacion
Pred : no sea heroderade su marido, le prefiercien la acusació

GT   : cion, y Capellan de su Magestad, y no he allado en el cosa contraria a
Pred : cion, y Capellan de su Magestad, y no he hallado en el cosa contraria a

GT   : villete al
Pred : villete al.

GT   : dre de Iacinto Gomez, para que diesse bienes de su
Pred : dre de Iacinto Gomez para que diesse bienes de su

GT   : los Corregidores, Assistente, Governado-
Pred : los Corregidores, Assistente, Governado-
```

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Saved best TRAIN loss model (stage2)
Saved best CER model (stage2)

Epoch 6/25
Train Loss : 0.2061
Val Loss   : 0.5421
CER        : 0.0796

Sample Predictions:

GT   : Iulius Clar. q. 58. num. 31. vers. contingit etiam. Bona-
Pred : Tulius C lat. q. 38. num. 3. ves si contingicetiam. Bona-

GT   : migo. Todo os sirva por bien, y aquí con desentrañable
Pred : migo Todo os strua por hien, y aqui con desontrañable

GT   : Escriuano, en 19. de Enero de 1640. por Iacinto
Pred : Eseruanos en Ió de Encro de1640., por Iacinto

GT   : no sea heredera de su marido, se prefiere en la acusacion
Pred : no sea leroderade su marido, le prefiercen la acusació

GT   : 12 Otra carta de pago, otorgada en esta Vi-
Pred :  121 Otra carta de pago, otorgada en esta Vi-
```

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alidation: 100%|████████████████████████████████████████████████████████████
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Saved best TRAIN loss model (stage2)
Saved best CER model (stage2)

Epoch 7/25
Train Loss : 0.1892
Val Loss   : 0.4475
CER        : 0.0767

Sample Predictions:

GT   : han escrito libros Criticos en nuestros tiempos, han procurado adornarlos con verdaderas, y doctas
Pred : han escrito libros Criticos en nuestros ciempos, han pre cuado adernarles cen ver dacteras, y cr etas

GT   : 12 Otra carta de pago, otorgada en esta Vi-
Pred :  121 Otra carta de pago, otorgada en esta Vi-

GT   : momento su remission, vt fere in terminis resoluune
Pred : momento su remission, ve fere in terminis resoluunc

GT   : oficio de la mesma manera: como el cuchillo; y las tixeras, se hizieron ambos para cortar, pero no
Pred : sficin de la mes na maneras comn el cichillo ; y las tixeras, setizieron amb o, para certar, yero no

GT   : de la casa, y llevandola a otras partes para usar mal de
Pred : de la casa, y llevandola ajotras partes para vlar mal de

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Saved best TRAIN loss model (stage2)

Epoch 8/25
Train Loss : 0.1742
Val Loss   : 0.4631
CER        : 0.0772

Sample Predictions:

GT   : Y de la pena que se deve imponer por los dichos de-
Pred : Y de la pena que se deue imponer por los dichos de

GT   : primer lugar en acusar, le tiene priuative en perdonar,
Pred : primer lugar en acusar, le tiene priuatiue en perdonar,I

GT   : el disseño, la luz, y el exemplo,
Pred : el disseño, la luz, y el exemplo,

GT   : to del lenguaje Latino, engaño en los venideros a muchos: y assi el bienavent
urado San Isidoro, que
Pred : to desenguage Latino, engaho en los venideros a mucstos yasi ci bienaueinitui
rado s a lsidoro, que

GT   : daron deuiendo, quando murio doña Beatriz
Pred : daron deuiendo, quando murio doña Beatriz

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Saved best CER model (stage2)

Epoch 9/25
Train Loss : 0.1745
Val Loss   : 0.4512
CER        : 0.0737

Sample Predictions:

GT   : migo. Todo os sirva por bien, y aquí con desentrañable
Pred : migo Todo os stru por bien, y aqui con desentrañable

GT   : y que se pidio su aprouacion, y que se auia dado
Pred : y que se pidio su aprouacion, y que se auia dado

GT   : Y de la pena que se deve imponer por los dichos de-
Pred : Y de la pena que se deue imponer por los dichos de

GT   : Santiago.
Pred : Santiago-

GT   : han escrito libros Criticos en nuestros tiempos, han procurado adornarlos con
verdaderas, y doctas
Pred : han escrito libros Criticos en nuestros ciempos, han pre cuado adernarles cen
ver daceras, y ce etas
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Saved best VALIDATION loss model (stage2)
Saved best CER model (stage2)

Epoch 10/25
Train Loss : 0.1745
Val Loss   : 0.3846
CER        : 0.0723

Sample Predictions:

GT   : cap penult.
Pred : tap penult-

GT   : Escriuano, en 19. de Enero de 1640. por Iacinto
Pred : Eseruano, en Ió de Encro de1640, por Iacinto

GT   : tonio ha de ser condenado a q avien
Pred : tonio ha de ser condenado a q avien

GT   : daron deuiendo, quando murio doña Beatriz
Pred : daron deuiendo, quando murio doña Beatriz

GT   : Matt. I8. proponeis por norma del can-
Pred : Matr. 18. proponeis por norma del can-

alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.23s/it]
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Saved best TRAIN loss model (stage2)
Saved best VALIDATION loss model (stage2)
Saved best CER model (stage2)

Epoch 11/25
Train Loss : 0.1527
Val Loss   : 0.3595
CER        : 0.0708

Sample Predictions:

GT   : primer lugar en acusar, le tiene priuative en perdonar,
Pred : primer lugar en acusar, le tiene priuatiue en perdonar,I

GT   : parece de sus palabras, que traducidas en castellano so-
Pred : pareee dessus palabras, que traducidas en Castellano so-

GT   : Iulius Clar. q. 58. num. 31. vers. contingit etiam. Bona-
Pred : Tulius C lact. q. 38. num. 3. ves siecontingic etiam. Bona-

GT   : muerte de su marido, e el mari
Pred : muerte de su marido, e el mariu

GT   : cion, y Capellan de su Magestad, y no he allado en el cosa contraria a
Pred : cion, y Capellan de su Magestad, y no he hallado en el cosa contraria a

alidation: 100%|████████████████████████████████████████████████████████
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Saved best CER model (stage2)

Epoch 12/25
Train Loss : 0.1564
Val Loss   : 0.4325
CER        : 0.0703

Sample Predictions:

GT   : conclusa la causa
Pred : conclusa la causa

GT   : Matt. I8. proponeis por norma del can-
Pred : Matr. 18. proponeis por norma del can-

GT   : no sea heredera de su marido, se prefiere en la acusacion
Pred : no sea herodera de su marido, se prefiercen la acusació

GT   : el disseño, la luz, y el exemplo,
Pred : el disseño, la luz, y el exemplo,

GT   : dre de Iacinto Gomez, para que diesse bienes de su
Pred : dre de Iacinto Gomez para que diesse bienes de su
```

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alidation: 100%|████████████████████████████████████████████████████████
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Saved best TRAIN loss model (stage2)
Saved best CER model (stage2)

Epoch 13/25
Train Loss : 0.1490
Val Loss   : 0.5332
CER        : 0.0683

Sample Predictions:

GT   : conclusa la causa
Pred : conclusa la causa

GT   : han escrito libros Criticos en nuestros tiempos, han procurado adornarlos con
verdaderas, y doctas
Pred : han escrito libros Criticos en nuestros ciempos, han pre curado adernarlos ce
n ver dacteras, y ce etas

GT   : el disseño, la luz, y el exemplo,
Pred : el disseño, la luz, y el ejemplo,

GT   : parece de sus palabras, que traducidas en castellano so-
Pred : paree e dessus palabras, que traducidas en Castellano so-

GT   : mandamiento para cobrar las penas pecuniarias, y
Pred : mandamiento para cobrar las penas pecuniarias, y

alidation: 100%|████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.12s/it]
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```
Saved best TRAIN loss model (stage2)
Saved best CER model (stage2)

Epoch 14/25
Train Loss : 0.1395
Val Loss   : 0.4535
CER        : 0.0654

Sample Predictions:

GT   : parte para el juez que lo sentenciare, y la
Pred : parte para el juez que lo sentenciare, y la

GT   : fee en publica forma como por corrector
Pred : fe e en publica forma como por corrector

GT   : daron deuiendo, quando murio doña Beatriz
Pred : daron deuiendo, quando murio doña Beatriz

GT   : han escrito libros Criticos en nuestros tiempos, han procurado adornarlos con
verdaderas, y doctas
Pred : han escrito libros Criticos en nuestros ciempos, han pre curado adernarlos ce
n ver dacteras, y ce etas

GT   : mandamiento para cobrar las penas pecuniarias, y
Pred : mandamiento para cobrar las penas pecuniarias, y

alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.14s/it]
Saved best CER model (stage2)

Epoch 15/25
Train Loss : 0.1481
Val Loss   : 0.4048
CER        : 0.0649

Sample Predictions:

GT   : dre de Iacinto Gomez, para que diesse bienes de su
Pred : dre de Iacinto Gomez para que diesse bienes de su

GT   : Iulius Clar. q. 58. num. 31. vers. contingit etiam. Bona-
Pred : Tulius C lac. q. 58. num. 3. ves si contingic etiam. Bona-

GT   : villete al
Pred : villete al.

GT   : tonio ha de ser condenado a q avien
Pred : tonio ha de ser condenado a q avien

GT   : Matt. I8. proponeis por norma del can-
Pred : Matt. 18. proponeis por norma del can-
```

alidation: 100%|██| 3/3 [00:12<00:00, 4.10s/it]
Saved best TRAIN loss model (stage2)

Epoch 16/25
Train Loss : 0.1376
Val Loss   : 0.3946
CER   : 0.0683

Sample Predictions:

GT   : parte para el juez que lo sentenciare, y la
Pred : parte para el juez que lo sentenciare, y la

GT   : tido; y parecidole mal, pues la acepcion del vocablo en su propia Etymologia es la verdadera signi-
Pred : rido, y parecidolemal, pues la acercion del vocablo en su propia Yeymoloziaes la vtrosderaseni

GT   : los que han deseado mucho cumplir con las obligaciones de aquel oficio, de cu yas obras no ten-
Pred : los que han deseado mucho cumplir con las obligaciones de aque l oñcio, ue cu yas obras no ten-

GT   : rradas las puertas con llaves, y usado de otras falsas, y
Pred : tradas las puertas con llaves, y vsado de otras falsas, y

GT   : muerte de su marido, e el mari
Pred : muerte de su marido, e el mariu

alidation: 100%|██| 3/3 [00:12<00:00, 4.14s/it]

```
Epoch 17/25
Train Loss : 0.1444
Val Loss   : 0.4724
CER        : 0.0654

Sample Predictions:

GT   : crianza de la niñez. Assi sea,
Pred : crianza de la Niñez. Assi sea,

GT   : de la casa, y llevandola a otras partes para usar mal de
Pred : de la casa, y llevandola aiotras partes para vsar mal de

GT   : rradas las puertas con llaves, y usado de otras falsas, y
Pred : tradas las puertas con llaves, y vsado de otras falsas, y

GT   : los que han deseado mucho cumplir con las obligaciones de aquel oficio, de cu
yas obras no ten-
Pred : los que han deseado mucho cumplir con las obligaciones de aque l oñcio, ue cu
yas obras no ten-

GT   : tonio ha de ser condenado a q avien
Pred : tonio ha de ser condenado a q avien

alidation: 100%|████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.16s/it]
Saved best TRAIN loss model (stage2)
Saved best CER model (stage2)

Epoch 18/25
Train Loss : 0.1372
Val Loss   : 0.3749
CER        : 0.0634

Sample Predictions:

GT   : to del lenguaje Latino, engaño en los venideros a muchos: y assi el bienavent
urado San Isidoro, que
Pred : to de senguage Latino, engaho en los venideros amucstos, y assi ci bienaventu
irado s u Isidoro, que

GT   : Santiago.
Pred : Santiago-

GT   : C
Pred :

GT   : parece de sus palabras, que traducidas en castellano so-
Pred : pareee de sus palabras, que traducidas en Castellano so-

GT   : y que se pidio su aprouacion, y que se auia dado
Pred : y que se pidio su aprouacion, y que se auia dado
```

```
alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.16s/it]
Saved best TRAIN loss model (stage2)

Epoch 19/25
Train Loss : 0.1247
Val Loss   : 0.3754
CER        : 0.0669

Sample Predictions:

GT   : Matt. I8. proponeis por norma del can-
Pred : Matt. 18. proponeis por norma del can-

GT   : muerte de su marido, e el mari
Pred : muerte de su marido, e el mariu

GT   : cion, y Capellan de su Magestad, y no he allado en el cosa contraria a
Pred : cion, y Capellan desu Magestad, y no he hallado en el cosa contraria a

GT   : migo. Todo os sirva por bien, y aquí con desentrañable
Pred : migo Todo os siru por bien y aquí con desentrañable

GT   : Santiago.
Pred : Santiago-

alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.17s/it]
```

```
Saved best VALIDATION loss model (stage2)
Saved best CER model (stage2)

Epoch 20/25
Train Loss : 0.1321
Val Loss   : 0.3585
CER        : 0.0624

Sample Predictions:

GT   : cap penult.
Pred : tap penult

GT   : gunas Etymologias que quiso deduzir de los vocablos que el fingio ser origina
riamente Latinos, no
Pred : guaas Eerntalogias que quiso de duzir de los vocablos que el singio ser origi
nariamente Latinos, no

GT   : Escriuano, en 19. de Enero de 1640. por Iacinto
Pred : Eseruano, en Ió de Encro de 1630 por Iacinto

GT   : y que se pidio su aprouacion, y que se auia dado
Pred : y que se pidio su aprouacion, y que se auia dado

GT   : los que han deseado mucho cumplir con las obligaciones de aquel oficio, de cu
yas obras no ten-
Pred : los que han deseado mucho cumplir con las obligaciones de aque l oncio, ue cu
vas obras no ten-


alidation: 100%|████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.15s/it]
```

```
Saved best CER model (stage2)

Epoch 21/25
Train Loss : 0.1368
Val Loss   : 0.4360
CER        : 0.0619

Sample Predictions:

GT   : oficio de la mesma manera: como el cuchillo; y las tixeras, se hizieron ambos
para cortar, pero no
Pred : sficin de la mesna maneras como el cnchillo ; y las tixeras, setizieron ambos
para certar, yero no

GT   : los Corregidores, Assistente, Governado-
Pred : los Corregidores, Assistente, Governado-

GT   : fee en publica forma como por corrector
Pred : fe e en publica forma como por corrector

GT   : testimonio de la particion, que se auia hecho de
Pred : testimonio de la patricion, que se avia hecho de

GT   : la cosa mas recebida que ay en los
Pred : la cosa mas recebida que ay en los


alidation: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.15s/it]
Epoch 22/25
Train Loss : 0.1318
Val Loss   : 0.3818
CER        : 0.0619

Sample Predictions:

GT   : tonio ha de ser condenado a q avien
Pred : tonio ha de ser condenado a q avien

GT   : villete al
Pred : villete al.

GT   : daron deuiendo, quando murio doña Beatriz
Pred : daron deuiendo, quando murio doña Beatriz

GT   : oficio de la mesma manera: como el cuchillo; y las tixeras, se hizieron ambos
para cortar, pero no
Pred : sficin de la mesna maneras como el cnchillo ; y las tixeras, setizieron ambos
para certar, yero no

GT   : parte que le toca de la dote de su madre, y tambien
Pred : parte que le toca de la dote de su madre, y tambi-
```

alidation: 100%|██| 3/3 [00:12<00:00, 4.14s/it]
Saved best TRAIN loss model (stage2)
Saved best VALIDATION loss model (stage2)
Saved best CER model (stage2)

Epoch 23/25
Train Loss : 0.1184
Val Loss   : 0.3526
CER   : 0.0610

Sample Predictions:

GT   : han escrito libros Criticos en nuestros tiempos, han procurado adornarlos con verdaderas, y doctas
Pred : han escrito libros Criticos en nuestros ciempos, han pre curado adernarlos ce n verdacteras, y ce das

GT   : C
Pred : .G

GT   : ca de Caycedo su hija, donzella noble y honesta, a
Pred : ta de Caycedo su hija, donzella noble y hon esta, a

GT   : to del lenguaje Latino, engaño en los venideros a muchos: y assi el bienavent urado San Isidoro, que
Pred : to de senguage Latino, engañio en los venideros amucsos, y assi ci bienaventu rado s un Isidoro, que

GT   : daron deuiendo, quando murio doña Beatriz
Pred : daron deviendo, quando murio doña Beatriz

alidation: 100%|██| 3/3 [00:12<00:00, 4.15s/it]

```
Saved best CER model (stage2)

Epoch 24/25
Train Loss : 0.1285
Val Loss   : 0.4388
CER        : 0.0605

Sample Predictions:

GT   : tonio ha de ser condenado a q avien
Pred : tonio ha de ser condenado a q avien

GT   : migo. Todo os sirva por bien, y aquí con desentrañable
Pred : migo Todo os sirua por bien, y aquí con desentrañable

GT   : conclusa la causa
Pred : conclusa la causa

GT   : villete al
Pred : villete al.

GT   : primer lugar en acusar, le tiene priuative en perdonar,
Pred : primer lugar en acusar, le tiene priuatiue en perdonar, a

alidation: 100%|████████████████████████████████████████████████████████
████| 3/3 [00:12<00:00,  4.15s/it]
Saved best TRAIN loss model (stage2)
Saved best CER model (stage2)

Epoch 25/25
Train Loss : 0.1044
Val Loss   : 0.4454
CER        : 0.0590

Sample Predictions:

GT   : MADRE, que se porten como sus
Pred : MADRE, que se porten como sus

GT   : de la casa, y llevandola a otras partes para usar mal de
Pred : de la casa, y llevandola aiotras partes para vsar mal de

GT   : Y de la pena que se deve imponer por los dichos de-
Pred : Y de la pena que se deue imponer por los dichos de

GT   : C
Pred :

GT   : gunas Etymologias que quiso deduzir de los vocablos que el fingio ser origina
riamente Latinos, no
Pred : guaas Ecrntalogias que quiso de duzir de sos vocablos que el singio ser brigi
nariamente Latinos, no
```

Full Model Training- Final Results:

The model was trained for 25 Epochs and the best **Character Error Rate** achieved on Validation Data was **5.9% in Epoch 25**.

Thus, the model predicted about 94% characters from the Validation dataset correctly and 5.9% of characters incorrectly.

In [55]:
```
#Loading best model:
model.load_state_dict(torch.load("stage2_best_cer_model.pth"))
```

Out[55]:
```
<All keys matched successfully>
```

In [56]:
```
#Downloading best model for later use:
FileLink("stage2_best_cer_model.pth") #Download link for only model parameters
```

Out[56]: stage2_best_cer_model.pth

In [57]:
```
#Saving additionmal necessary information:
torch.save({
    "model_state_dict": model.state_dict(),
    "char_to_idx": char_to_idx,
    "idx_to_char": idx_to_char
}, "stage2_best_cer_model_complete1.pth")

#Downoad link for full model:
FileLink("stage2_best_cer_model_complete1.pth")
```

Out[57]: stage2_best_cer_model_complete1.pth

In [58]:
```
#On kernel dying, code to use saved parameters:
#model = Model(opt)     # load same architecture as training- this line, then replc
model.load_state_dict(torch.load(r"C:\Users\91861\Downloads\stage2_best_cer_model.p
```

Out[58]:
```
<All keys matched successfully>
```

Stage 3-Output refining using LLM Integration:

Step 8- Integrating a Large Language Model:

The Goal: Use a trained large language model to refine the predictions of the CRNN model.

Additional idea: Provide a few pairs of ground truth labels and OCR predictions from the train dataset to an LLM. Prompt the LLM to analyze the patterns and differrences between the predictions and the ground truth and remember this information .Next, add the OCR predictions, one line at a time, to the prompt and ask the LLM Model to predict the ground truth from this predicted OCR output. Try some differrent prompting patterns and select the

one with the lowest CER as the final prompt.Evaluate this best prompt on the test data to get the final accuracy score.

This is method is known as **'Few Shot Prompt Tuning'**.

To integrate an LLM-based post-processing step into the OCR pipeline, local language models were deployed using the Ollama framework. Relying on cloud APIs such as Google Gemini can introduce rate limits and usage restrictions, which may hinder large-scale batch processing of OCR outputs. To avoid these constraints and ensure uninterrupted processing, Ollama was used to run open-weight language models directly on local hardware, enabling efficient and unrestricted local inference.

Models Used-

1. Mistral 7B
2. Qwen 2.5 7B

In [52]:
```
#Randomly sampling (OCR Predictions,Ground truth label) from training data to input
model.eval()

all_preds = []
all_labels = []

with torch.no_grad():
    for images, labels in tqdm(train_loader, desc="Running OCR on train data"):
        #images = images.to(device)
        preds = model(images, None)
        decoded_preds = greedy_decode(preds, idx_to_char)
        all_preds.extend(decoded_preds)
        all_labels.extend(labels)

# Pairing predictions with GT labels:
pairs = list(zip(all_preds, all_labels))

# Taking a sample of 20 random examples:
sample_pairs = random.sample(pairs, min(20, len(pairs)))
```

```
unning OCR on train data: 100%|████████████████████████████████████████████████
██| 47/47 [03:27<00:00,  4.41s/it]
```

First, the Mistral-7B model is used due to its strong performance relative to its size and its reputation for efficient inference. As a lightweight yet capable model, it provides a practical starting point for experimentation with LLM-based OCR correction while maintaining manageable computational requirements.

Passing OCR Predictions to LLM 'Mistral-7B'- Using a simple prompt (without examples) to set a baseline.

In [63]:
```
def llm_refine_line(line):
    prompt = f"""The given line was extracted from a 16th century Spanish manuscrip
Your goal is to correct OCR errors and return a corrected line.
```

```
Rules:
- Correct spelling mistakes. While doing so, preserve historical spelling.
- Do NOT modernize language.
- Do NOT explain anything- JUST MAKE THE CHANGES.DONOT EXPLAIN ANYTHING ABOUT THE C
- Do NOT add notes like the following- Explanation: I've only made spelling correct
- Do NOT add notes like-"no changes made"
- Do NOT add notes like-(preserving the number as it is, but adding the correct pun
-  Do NOT add notes like-(assuming "pitan" was meant to say "pito", which means "wh
- Do NOT add words.
- Return ONLY the corrected line.
- If there is a number, Do NOT change it.
- Do NOT add accents if the input line does not have it.
- Do NOT capitalize a letter if the input line letter is not capitalized and vice v
- Do NOT make changes like- replacing v with 'uno'


The line which is to be corrected: {line}

Corrected:
"""
    response = ollama.chat(
        model="mistral",
        messages=[{
            "role": "user",
            "content": prompt}])

    output = response['message']['content'].strip()

    # Sometimes models add prefixes like "Corrected:" Thus,removing the string:
    output = output.replace("Corrected:", "").strip()

    return output
```

In [53]:
```
#Getting first 100 training data points to input to LLM:
all_preds=all_preds[:100]
all_labels=all_labels[:100]
```

In [64]:
```
corrected_preds = [] #list to store LLM results.

for line in all_preds:
    corrected = llm_refine_line(line)
    corrected_preds.append(corrected)
```

In [65]:
```
print("\nExample LLM corrections:\n")

for i in range(10):
    print("OCR :", all_preds[i])
    print("LLM :", corrected_preds[i])
    print("GT  :", all_labels[i])
    print()
```

```
Example LLM corrections:

OCR : y primero el dicho libro este corregido y
LLM : y primero el dicho libro este correjido y
GT  : y primero el dicho libro este corregido y

OCR : sieio de la anes ma moneae eomo elcuehisto, y las tiver ano del lales enan lo,
pira cnr earor ero n
LLM : seigio de la anses monesae oelo elcuehisto, y las tiver ano del lalces enan l
o, pirra conear oro n
GT  : oficio de la mesma manera: como el cuchillo; y las tixeras, se hizieron ambos
para cortar, pero no

OCR : sino también de assis- 46
LLM : sino también de asís- 46
GT  : sino también de assis- 46

OCR : co años jura la relacion de esta escritura, y estar, y
LLM : có annos jura la relación de esta escritura, y estaré, y
GT  : co años jura la relacion de esta escritura, y estar, y

OCR : ian por milagro en ellas, por el grande conocimiento, y nunca visto en hombre
Español de la
LLM : Yan por milagro en ellas, por el gran conocimiento, y nunca visto en homem Esp
añol de la
GT  : nian por milagro en ellas, por el grande conocimiento, y nunca visto en hombre
Español de la

OCR : arras de su madre, a mayor abudamiento las aprue-
LLM : arras de su madre, a mayordabidamiento las aprueba
GT  : arras de su madre, á mayor abundamiento las aprue-

OCR : de colusion, como lo muestra la diligencia, y
LLM : de colisión, cómo lo muestra la diligencia, y
GT  : de colusion, como lo muestra la diligencia, y per-

OCR : del Numero desta villa, entre lacinto Gomez Pi-
LLM : del número desta villa, entre Lacinto Gómez Pí
GT  : del Numero desta villa, entre Iacinto Gomez Pi-

OCR : do, y por sentencia de reuista se le conmutó la di-
LLM : do, y por sentencia de reyasta se le comutó la di-
GT  : do, y por sentencia de reuista se le conmutó la di-

OCR : ble post Bart. &. Bald. Nicol. Boer. decis. l,: la num. 2.
LLM : El libro poste Barrt. y Bald. Nicol. Boer. decidis. I: la num. 2.
GT  : ble, post Bart. & Bald. Nicol. Boer. decis. 233. a num. 2.
```

In [66]:

```python
#Calculating the CER and WER values after passing OCR output through LLM

cer_before = calculate_cer(all_preds, all_labels)
cer_after = calculate_cer(corrected_preds, all_labels)

wer_before = calculate_wer(all_preds, all_labels)
wer_after = calculate_wer(corrected_preds, all_labels)
```

```
print("CER before LLM:", cer_before)
print("CER after LLM :", cer_after)

print("WER before LLM:", wer_before)
print("WER after LLM :", wer_after)
```

```
CER before LLM: 0.09545913218970736
CER after LLM : 0.19455095862764885
WER before LLM: 0.28632938643702904
WER after LLM : 0.4133476856835307
```

Currently, out of all the iterations with different prompts using **Mistral 7B** (all with no examples provided), the best iteration achieved 19% CER error rate which is much higher than the error rate of the simple CRNN model which is 9.5% on training data. It appears that the model is generating text in some cases instead of only editing, when required, according to rules. Next, we use Model Qwen 2.5 7B.

Passing OCR Predictions to LLM Model Qwen 2.5 7B- Using a simple prompt (without examples) to set a baseline. This model was selected for experimentation as it provides a strong balance between performance and computational efficiency.It also offers improved instruction-following capability compared to several other models of similar size.

In [101…

```
def refine_line(line, sample_pairs):

    prompt = f"""The given line was extracted from a 16th century Spanish manuscrip
Your goal is to correct OCR errors and return a corrected line.

Rules:
- Correct spelling mistakes. While doing so, preserve historical spelling.
- Do NOT modernize language.
- Do NOT explain anything- JUST MAKE THE CHANGES.DONOT EXPLAIN ANYTHING ABOUT THE C
- Do NOT add notes like the following- Explanation: I've only made spelling correct
- Do NOT add notes like-"no changes made"
- Do NOT add notes like-"preserving the number as it is, but adding the correct pun
-  Do NOT add notes like-"assuming "pitan" was meant to say "pito", which means "wh
- Do NOT add words.
- Return ONLY the corrected line.
- If there is a number, Do NOT remove it.
- Do NOT add accents if the input line does not have it.
- Do NOT capitalize a letter if the input line letter is not capitalized and vice v
- Do NOT make changes like- replacing v with 'uno'
- VERY IMPORTANT- If a word is incomplete followed by a "-" sign, do NOT complete t


The line which is to be corrected: {line}
Return EXACTLY one line of text.
No explanation.
No extra characters.
THIS IS A TEXT EDITING TASK(ONLY WHEN REQUIRED) AND NOT A TEXT GENERATION TASK.
Corrected:
"""

    response = ollama.chat(
```

```
        model="qwen2.5:7b",
        messages=[{"role": "user", "content": prompt}],
        options={
            "temperature": 0,
            "top_p": 1,
    "repeat_penalty": 1.1})

    return response["message"]["content"].strip()
```

In [106…

```
corrected_preds = [] #list to store LLM results.

for line in all_preds:
    corrected = llm_refine_line(line)
    corrected_preds.append(corrected)
```

In [103…

```
print("\nExample LLM corrections:\n")

for i in range(10,100):
    print("OCR :", all_preds[i])
    print("LLM :", corrected_preds[i])
    print("GT  :", all_labels[i])
    print()
```

Example LLM corrections:

OCR : pragmatica por nos hecha sobre la impres
LLM : pragmatica por nos hecha sobre la impres
GT : pragmatica por nos hecha sobre la impres

OCR : cap
LLM : cap

(no changes are needed)
GT : cap

OCR : los vucablos desengu ñan de dos earores muy assena dos en la igrorancia de los icrsla sos,s
LLM : los vucablos desengu nan de dos errores muy asenados dos en la igrorancia de l os islas sos
GT : de los vocablos, desengañan de dos errores muy assentados en la ignorancia de los iergllages, pro-

OCR : matro abrieron ema veatana en el dicho aposento d
LLM : matro abrieron emata veatana en el dicho aposento d
GT : cuatro abrieron una ventana en el dicho aposento de

OCR : Escriuano, en 19. de Enero de. 1640, por Iacinto
LLM : Escriuano, en 19. de Enero de. 1640, por Iacinto
GT : Escriuano, en 19. de Enero de 1640. por Iacinto

OCR : rem relinquat illa
LLM : rem relinquat illa
GT : rem relinquat illa

OCR : conio ha de ser condenado a q avien
LLM : conio ha de ser condenado a que aviene
GT : tonio ha de ser condenado a q avien

OCR : nuestra cedula, licencia y privilegio, y la
LLM : nuestra cedula, licencia y privilegio, y la
GT : nuestra cedula, licencia y privilegio, y la

OCR : tam, por no poder concurrir con la viuda, como por
LLM : tam, por no poder concurir con la viuda, como por
GT : tam, por no poder concurrir con la viuda, como por

OCR : setur propinquior persona ad atcusandm. 6 .ranfigendi-
LLM : setur propinquior persona ad atcusandm. 6 .ranfigendo-
GT : setur propinquior persona ad accusandu, & transigendu;

OCR : Synonymms, penicndo rodos losrocablos que dia en sgnisiear una cos useia lendo ciuto, que ni
LLM : Synonymms, penicilindo rosas los loblablos que dia en signisier una cosa usari a leño ciudad, que ni
GT : Synonymos, poniendo todos los vocablos que dizen significar una cosa misma, si endo cierto, que ni

OCR : se con el matador solicita su causa, no es de momento,
LLM : se con el matador solicita su causa, no es de momento

GT : se con el matador solicita su causa, no es de momento;

OCR : para cortar
LLM : para cortar
GT : para cortar,

OCR : miento de las fry nolegias demde de serami parecer el mayor gulio de los burno
s ingenios, cemo
LLM : miento de las fry nolegias demonstra de serami parecer el mayor gulio de los b
urnos ingenios, ceomo
GT : miento de las Etymologias, demas de ser a mi parecer el mayor gusto de los bue
nos ingenios, como

OCR : num. 153.
LLM : num. 153.
GT : num. 153.

OCR : primir porel tiempo q su essemos sertuido, o
LLM : primir por el tiempo que su ese mos sertuido, o
GT : primir por el tiempo q fuessemos servido, o

OCR : cho tiempo persona alguna sin vuestra li-
LLM : cho tiempo persona alguna sin vuestra li-
GT : cho tiempo persona alguna sin vuestra li-

OCR : cibido por cuenta de su legitima 8000 reales por
LLM : cibido por cuenta de su legitima 8000 reales por
GT : cibido por cuenta de su legitima 8 000. reales por

OCR : gu Guinga que no sirmn dr otra cosa, sino de causar visa a los que la sibrn co
mo de ie que el voco-
LLM : gu Guinga que no sirmn d'otra cosa, sino de causar visao a los que la sibrn co
mo de ié que el voco-
GT : gua Griega, que no sirven de otra cosa, sino de causar risa a los que la sabe
n, como dezir que el voca-

OCR : cuya elegancia, y suiaeaa de senriod este liiro de vam, porque no puede conoce
rsa sanctasonn
LLM : cuiya elegancia, y suaheza de senyor este liiro de vos, porque no puede conoce
rsa sanctionon
GT : cuya elegancia, y sutileza descubrira este libro de v. m. porque no puede cono
cerse la metafora, o

OCR : de acosadores de la muerte con laviu
LLM : de acosadores de la muerte con la viu
GT : de acosadores de la muerte con la viu-

OCR : Y pues en la celestial Jeru-
LLM : Y pues en la celestial Jerusalem
GT : Y pues en la celestial Jeru-

OCR : derecho, aun enterminos de concurso de
LLM : derecho, aun entre términos de concurso de
GT : derecho, aun en terminos de concurso de

OCR : vor s andar en estas cosas sin ser notada, fuera dela po

LLM : vor s andar en estas cosas sin ser notada, fuera de la po
GT : por si andar en estas cosas sin ser notada, fuera de la po-

OCR : ntus rumolsias mezclo muchos de los vocablos la tines, dio ocasona le arisen d
e los nuta
LLM : ntus rumolsias mezclo muchos de los vocablos la tines, dio ocasion a la arisen
de los nuta
GT : en sus Etymologias mezclo muchos de los vocablos Latinos, dio ocasión a al irr
isiori de los no tan

OCR : que esto en vuestra infinita Sabi-
LLM : que esto en vuestra infinita Sabid
GT : que esto en vuestra infinita Sabi-

OCR : ciuil,
LLM : ciuil,
GT : civil,

OCR : pitan
LLM : piton
GT : pitan

OCR : ne officij vel alias los goza la muger, y no los hijos,
LLM : ne officij vel alias los goza la muger, y no los hijos,
GT : ne officij, vel alias, los goza la muger, y no los hijos, I.

OCR : D. Balthasar de Bastero y lledo, Obispo
LLM : D. Balthasar de Bastero y Ildeo, Obispo
GT : D. Balthasar de Bastero y lledo, Obispo

OCR : Por testimonio de Antonio Cadenas Escri
LLM : Por testimonio de Antonio Cadena Escri
GT : 6 Por testimonio de Antonio Cadenas Escri-

OCR : ea crcor que ayabos, d mas vocoblos, que siegnlfiquem una cosa mismu, y este e
l a maadeetdo que e
LLM : ea corro que ayabos, d más vocablos, que se ignifiquen una cosa misma, y este
el amado que es
GT : es creer que ay dos, o mas vocablos, que sifnifiquen una cosa misma, y este es
ta mas assentado que el

OCR : sado puedan concurrir
LLM : sado puedan concurir
GT : sado) puedan concurrir

OCR : se porquesinui st,temor de lo que yo digo no os hu
LLM : se porque sin ui st, temor de lo que yo digo no os huy
GT : so: porque si….. temor de lo que yo digo, no os hu-

OCR : y merced vos damos licencia y facultad pa
LLM : y merced vos damos licencia y facultad pa
GT : y merced vos damos licencia y facultad pa

OCR : dotiu, & quae post antiquos tradie Menoch de arbitra-
LLM : dotiu, & quae post antiquos traditionem Menoch de arbitrario-
GT : dotium, & quae post antiquos tradit Menoch. de arbitra-

OCR : nelia, & §.item lex lulia, init. de publicis iudicijs
LLM : nelia, & §. item lex Lulia, init. de publicis iudicijs
GT : nelia, & §. item lex Iulia, int. de publicis iudicijs.

OCR : he visto un Librito, cuyo titulo es: Ins-
LLM : he visto un Librito, cuyo titulo es: Insi-
GT : He visto un Librito, cuyo titulo es: Ins-

OCR : .Y en quanto a la hija, que primero represento tan
LLM : .Y en quanto a la hija, que primo represento tan
GT : Y en quanto a la hija, que primero represento tan

OCR : Tribunales deste Reyno, y se prueua ab ordine literae in
LLM : Tribunales deste Reyno, y se prueua ab ordine literae in
GT : Tribunales deste Reyno, y se prueva ab ordine literae in

OCR : texados, y otras sacandola de su casa, y llevandola a o-
LLM : texados, y otras sacandola de su casa, y llevandola a otro
GT : texados, y otras sacandola de su casa, y llevandola a o-

OCR : ,de los señores Alcaldes, en que man-
LLM : ,de los señores Alcaldes, en que man-
GT : de los señores Alcaldes, en que man-

OCR : LICENCIADO DON BALTASAR
LLM : LICENCIADO DON BALTASAR
GT : LICENCIADO DON BALTASAR

OCR : enseñar a los de nuestro siglo a ser estudiantes como lo han de ses, pues en e
l golse de los regecios,
LLM : enseñar a los de nuestro siglo a ser estudiantes como lo han de ses, pues en e
l golse de los reyes,
GT : enseñar a los de nuestro siglo a ser estudiantes como lo han de ser, pues en e
l golfo de los negocios,

OCR : platillos, sinonos gouernamos con prudencia, lacame y
LLM : platillos, sinonos governamos con prudencia, lacame y
GT : platillos si no nos gobernamos con prudencia. Jacome y,

OCR : el dicho original para q se vea si la dicha
LLM : el dicho original para que se vea si la dicha
GT : el dicho original para q se vea si la dicha

OCR : que escrivió la adultera a su amante, de quibus in proces-
LLM : que escribio la adultera a su amante, de quibus en proceso-
GT : que escribió la adultera a su amante de quibus in proces-

OCR : quien les aparta de Vos, y les
LLM : quien les aparta de Vos y les
GT : quien les aparta de Vos, y les

OCR : neos porque como por el matrimonio se hizieron am-
LLM : neos porque como por el matrimonio se hicieron amigo
GT : neos porque como por el matrimonio se hizieron am-

OCR : yor gloria, Amen.
LLM : yor gloria, Amen.
GT : yor gloria. Amen.

OCR : varon, unico sucessor en el mayorazgo de su padre, mo-
LLM : varon, unico sucessor en el mayorazgo de su padre, mo-
GT : varon, unico sucessor en el mayorazdo de su padre, mo-

OCR : cosa in suis communibus, vers. pax uxoricidae, lib. 1. Pe
LLM : cosa in suis communibus, vers. pax uxoris, lib. 1. Pe
GT : cosa in suis communibus, vers. pax vxoricidae, lib. I. Pe-

OCR : como queda fundado, nu 4. Lo segundo, porque el hijo
LLM : como queda fundado, nu 4. Lo segundo, porque el hijo
GT : como queda fundado, nu. 4. Lo segundo, porque el hijo

OCR : hijo todo loque montava su legitima marerna,
LLM : hijo todo lo que montaba su legitima marera
GT : hijo todo lo que montaua su legitima materna,

OCR : dicaron, y pagaron en diferentes deuitos q. se que
LLM : dicaron, y pagaron en diferentes deudas que
GT : dicaron, y pagaron en diferentes deuitos que se que-

OCR : gunas Eeymologias que quiso devuzir de lus vocablos que el singio ser originar iamente latinos, no
LLM : gunas Eeymologias que quiso devuzir de los vocablos que el singular sea origin ariamente latinos, no
GT : gunas Etymologias que quiso deduzir de los vocablos que el fingio ser originar iamente Latinos, no

OCR : REY
LLM : REY
GT : REY

OCR : truccion de Christiana, y Politica Cortesa-
LLM : truccion de Cristiana, y Política Cortésa-
GT : truccion de Christiana, y Politica Cortesa-

OCR : e procede en los demas delitos de que es acusado don-
LLM : e procede en los demas delitos de que es acusado don
GT : te procede en los demas delitos de que es acusado don

OCR : cacr
LLM : carg
GT : cacr

OCR : Oran,
LLM : Oran,
GT : Oran.

OCR : quedess obrase ha de leguir gran vtitnad, y honor a lanacion Española, esey my cobe nto de
LLM : quedess obrase ha de leguir gran virtud, y honor a la nacion Española, esey mo y cobra no de
GT : que desta obra se ha de seguir gran utilidad, y honor a la nacion Española, es

toy muy contento de

OCR : Syndal de losohispador de Gerona, vie-
LLM : Syndal de losohispaldor de Gerona, vie-
GT : Synodal de los Obispados de Gerona, Ur-

OCR : impresion esta conforme a el, y traygays
LLM : impresion esta conforme a él, y traygays
GT : impression esta conforme a el, y traygays

OCR : puede la muger suplir la omission del marido, ex Fa-
LLM : pueda la mujer suplir la omisión del marido, ex Fa-
GT : puede la muger suplir la omission del marido, ex Fa-

OCR : a cobranea se gasta-
LLM : a cobranea se gasta-
GT : a cobranza se gasta-

OCR : parte para el juez que lo sentenciare, y la
LLM : parte para el juez que lo sentenciare, y la
GT : parte para el juez que lo sentenciare, y la

OCR : duria fue soberana dignacion, y
LLM : duria fue sobrenada dignacion, y
GT : duria fue soberana dignacion, y

OCR : 4 En execución desta executoria, se despachó
LLM : 4 En execucion desta executoria, se despacho
GT : 4 En execución desta executoria, se despachó

OCR : PretenDE Antonio GomezPi
LLM : PretenDE Antonio GomezPi
GT : Pretende Antonio Gomez Pi

OCR : se le ha de satisfacer por los tres herederos, cada v-
LLM : se le ha de satisfacer por los tres herederos, cada ve
GT : se le ha de satisfacer por los tres herederos, cada v-

OCR : muerte dosu marido, queces la inspeccion que hizo s,
LLM : muerte dos marido, queces la inspeccion que hizo s,
GT : muerte de su marido, que es la inspreccion que hizo el derecho, aun en termino
s de concurso de acusadores,

OCR : da su felizidad en la acertada
LLM : da su felicidad en la acertada
GT : da su felizidad en la acertada

OCR : prenda, son estas seboru para queyo essimne tal deudo, quando ve m. por su per
sena no me diera
LLM : prenda, son estas seboru para que yo este imne tal deudo, cuando ve m. por su
persona no me da
GT : Prendas son estas señor, para que yo estime tal deudo, quando v. m. por su per
sona no me diera

OCR : Loprimero, quela viud tutn col y de mas agravios
LLM : Loprimero, que la viuda tuvo col y de más agravios

GT : Lo primero, que la viuda tiene el primer lugar en la acusacion de la muerte de
l marido, y demas agravios

OCR : buena cuenta a su padre dellas, ó pagar el precio
LLM : buena cuenta a su padre de ellas, ó pagar el precio
GT : buena cuenta a su padre dellas, ó pagar el precio

OCR : des y aparejos que de los dichos libros tu
LLM : des y aparejos que de los dichos libros tu
GT : des y aparejos que de los dichos libros tu

OCR : ta, con que los mil ducados fuessen ochocientos,
LLM : ta, con que los mil ducados fueresen ochocientos,
GT : ta, con que los mil ducados fuessen ochocientos,

OCR : fue fecha relacion q vos
LLM : fue fecha relacion q vos
GT : fue fecha relacion q vos

OCR : al fin de Pedro del Marmol nuestro escri-
LLM : al fin de Pedro del Marmol nuestro escribi
GT : al fin de Pedro del Marmol nuestro escri-

OCR : propiedad, pureza, y elegancia de una lengua se escriva en el tiempo que
LLM : propiedad, pureza, y elegancia de una lengua se escribia en el tiempo que
GT : propiedad, pureza, y elegancia de una lengua se escriva en el tiempo que

OCR : en su muger, por dos razones.
LLM : en su mujer, por dos razones.
GT : en su muger, por dos razones.

OCR : politica, vatones muy graves, y doctos, y por ser conueniente que de l
LLM : politica, vatones muy graves, y doctos, y por ser conveniente que de la
GT : politica, varones muy graves, y doctos, y por ser conveniente que de la

OCR : do contra Iacinto Gomez Pimentel,
LLM : do contra Iacinto Gomez Pimentell,
GT : do contra Iacinto Gomez Pimentel,

OCR : si sabe la muerte de su padre recono-
LLM : si sabia la muerte de su padre recono-
GT : si sabe la muerte de su padre recono-

OCR : antos años que trabaja, y ha comiengado a imprimir, porque tenendo yo opinion
d
LLM : antos años que trabaja, y ha comenzado a imprimir, porque teniendo yo opinión
d
GT : tantos años que trabaja, y ha comenzado a imprimir: porque teniendo yo opinion
de

OCR : buenas costumbres, sino que muy atento
LLM : buenos costumbres, sino que muy atento
GT : buenas costumbres, sino que muy atento

OCR : varón ausente no aueracusado, y la hija no pedir a su ma-
LLM : varón ausente no aueracusado, y la hija no pedir a su ma-

```
GT  : varon ausente no aver acusado, y la hija no perdir a su ma-

OCR : que os merecio tan regalados ca-
LLM : que os mereció tan regalados ca-
GT  : que os merecio tan regalados ca- 16.

OCR : carne una itaqueiam non suur duo,seduna caro, &cita
LLM : carne una itaqueiam non sur duo,seduna caro, &cita
GT  : carne una, itaque iam non sune duo, fedunacaro, & ita
```

In [104…

```python
#Calculating the CER and WER values after passing OCR output through LLM

cer_before = calculate_cer(all_preds, all_labels)
cer_after = calculate_cer(corrected_preds, all_labels)

wer_before = calculate_wer(all_preds, all_labels)
wer_after = calculate_wer(corrected_preds, all_labels)

print("CER before LLM:", cer_before)
print("CER after LLM :", cer_after)

print("WER before LLM:", wer_before)
print("WER after LLM :", wer_after)
```

```
CER before LLM: 0.09545913218970736
CER after LLM : 0.13299697275479314
WER before LLM: 0.28632938643702904
WER after LLM : 0.3261571582346609
```

Thus, Qwen 2.5 7B performs better than Mistral- the CER for Mistral was 19% while for Qwen it is 13.2%. However, it is still higher than simple CRNN. Next, we add some examples to the prompt to improve the Error Rate.

Few Shot Prompting: Adding examples of OCR output and ground truth labels to the Best prompt & Model (Qwen 2.5 7B) combination from the previous step :

In [116…

```python
def refine_line(line, sample_pairs):

    prompt = f"""The given line was extracted from a 16th century Spanish manuscrip
Your goal is to correct OCR errors and return a corrected line.

Rules:
- Correct spelling mistakes. While doing so, preserve historical spelling.
- Do NOT modernize language.
- Do NOT explain anything- JUST MAKE THE CHANGES.DONOT EXPLAIN ANYTHING ABOUT THE C
- Do NOT add notes like the following- Explanation: I've only made spelling correct
- Do NOT add notes like-"no changes made" or "(preserving original spelling)"- THIS
- Do NOT add notes like-"preserving the number as it is, but adding the correct pun
- Do NOT add notes like-"assuming "pitan" was meant to say "pito", which means "whi
- Do NOT add words.
- Return ONLY the corrected line.
- If there is ANY NUMBER, Do NOT remove it.Keep it exactly the same.
- Do NOT add accents if the input line does not have it.
- Do NOT capitalize a letter if the input line letter is not capitalized and vice v
```

```
- Do NOT make changes like- replacing v with 'uno'
- VERY IMPORTANT- If a word is incomplete followed by a "-" sign, do NOT complete t


The line which is to be corrected: {line}
Return EXACTLY one line of text.
No explanation.
No extra characters.
THIS IS A TEXT EDITING TASK(ONLY WHEN REQUIRED) AND NOT A TEXT GENERATION TASK.

Examples of input-output pairs are provided below:
{sample_pairs}

Each pair shows an OCR input and its correct output.
Analyze the patterns used to correct the OCR errors and apply similar corrections t

Corrected:
"""

    response = ollama.chat(
        model="qwen2.5:7b",
        messages=[{"role": "user", "content": prompt}],
        options={
            "temperature": 0,
            "top_p": 1,
    "repeat_penalty": 1.1})

    return response["message"]["content"].strip()
```

In [115…
```
sample_pairs=sample_pairs[:10] #Examples given to LLM
sample_pairs
```

Out[115…
```
[('nras Audiencias Alcaldes, alguaziles de la',
  'nras Audiencias Alcaldes, alguaziles de la'),
 ('mentel se reuoque el auto de la Sala',
  'mentel se reuoque el auto de la Sala'),
 ('ña Leonor Rauasco', 'ña Leonor Rauasco'),
 ('varon, unico sucessor en el mayorazgo de su padre, mo-',
  'varon, unico sucessor en el mayorazdo de su padre, mo-'),
 ('se que no contencara a vom, pero con todo eslo es de tan grande rtilidad el',
  'se que no contentara a v. m. pero con todo esso es de tan grande utilidad el'),
 ('re.', 're.'),
 ('bos una misma carne y sangre, no ay persona mas con-',
  'bos una misma carne y sangre, no ay persona mas con-'),
 ('gel, y Barcelona, Oc', 'gel, y Barcelona, Oc.'),
 ('impresion esta conforme a el, y traygays',
  'impression esta conforme a el, y traygays'),
 ('Señor.', 'Señor.')]
```

In [117…
```
corrected_preds = [] #list to store LLM results.

for line in all_preds:
    corrected = llm_refine_line(line)
    corrected_preds.append(corrected)
```

In [118…

```
print("\nExample LLM corrections:\n")

for i in range(10):
    print("OCR :", all_preds[i])
    print("LLM :", corrected_preds[i])
    print("GT  :", all_labels[i])
    print()
```

```
Example LLM corrections:

OCR : y primero el dicho libro este corregido y
LLM : y primero el dicho libro este corregido y
GT  : y primero el dicho libro este corregido y

OCR : sieio de la anes ma moneae eomo elcuehisto, y las tiver ano del lales enan lo,
pira cnr earor ero n
LLM : sieio de la añes mã monéae eemo elcuehisto, y las tiver ano del lales enan lo,
pirar coner ero n
GT  : oficio de la mesma manera: como el cuchillo; y las tixeras, se hizieron ambos
para cortar, pero no

OCR : sino también de assis- 46
LLM : sino también de asist- 46
GT  : sino también de assis- 46

OCR : co años jura la relacion de esta escritura, y estar, y
LLM : co años jura la relacion de esta escritura, y estar, y
GT  : co años jura la relacion de esta escritura, y estar, y

OCR : ian por milagro en ellas, por el grande conocimiento, y nunca visto en hombre
Español de la
LLM : ian por milagro en ellas, por el gran conocimiento, y nunca visto en hombre es
pañol de la
GT  : nian por milagro en ellas, por el grande conocimiento, y nunca visto en hombre
Español de la

OCR : arras de su madre, a mayor abudamiento las aprue-
LLM : arras de su madre, a mayordomía las aprueba
GT  : arras de su madre, á mayor abundamiento las aprue-

OCR : de colusion, como lo muestra la diligencia, y
LLM : de colusión, como lo muestra la diligencia, y
GT  : de colusion, como lo muestra la diligencia, y per-

OCR : del Numero desta villa, entre lacinto Gomez Pi-
LLM : del Numero desta villa, entre lacinto Gomez Pi-

No changes made.
GT  : del Numero desta villa, entre Iacinto Gomez Pi-

OCR : do, y por sentencia de reuista se le conmutó la di-
LLM : do, y por sentencia de reuisita se le conmuto la di-
GT  : do, y por sentencia de reuista se le conmutó la di-

OCR : ble post Bart. &. Bald. Nicol. Boer. decis. l,: la num. 2.
LLM : ble post Bart. &. Bald. Nicol. Boer. decis. l,: la num. 2.
GT  : ble, post Bart. & Bald. Nicol. Boer. decis. 233. a num. 2.
```

In [119…

```python
#Calculating the CER and WER values after passing OCR output through LLM

cer_before = calculate_cer(all_preds, all_labels)
cer_after = calculate_cer(corrected_preds, all_labels)
```

```
wer_before = calculate_wer(all_preds, all_labels)
wer_after = calculate_wer(corrected_preds, all_labels)

print("CER before LLM:", cer_before)
print("CER after LLM :", cer_after)

print("WER before LLM:", wer_before)
print("WER after LLM :", wer_after)
```

```
CER before LLM: 0.09545913218970736
CER after LLM : 0.15176589303733604
WER before LLM: 0.28632938643702904
WER after LLM : 0.3519913885898816
```

The best CER Error rate currently achieved on Passing OCR predictions of Training Data through a Large Language Model is 13.2%. (No guiding examples were provided and Qwen 2.5 7B Model was used in the instance corrosponding to this score).

So, in the current framework, applying LLM-based post-processing to the CRNN outputs is not improving overall performance. This suggests that the current LLM refinement approach implemented requires further redesign and optimization.

Next, I am proceeding with evaluating the performance of the CRNN model on Test Data.

Step 9- Evaluating Model on Test Dataset:

In [59]:
```
#Test data loader:
test_ds = OCRDataset(test_data, transform=None)

test_loader = DataLoader(
    test_ds,
    batch_size=16,
    shuffle=False,
    num_workers=0,
    collate_fn=ocr_collate)
```

In [61]:
```
#Loading best model:
model = Model(opt)

#Replacing final Layer:
model.Prediction = nn.Linear(
    model.Prediction.in_features,
    num_classes # =79(size of output from prediction layer)
)

model.load_state_dict(torch.load("stage2_best_cer_model.pth", #map_location=device
                                 ))
model.eval()
```

```
No Transformation module specified
```

Out[61]:
```
Model(
  (FeatureExtraction): ResNet_FeatureExtractor(
    (ConvNet): ResNet(
      (conv0_1): Conv2d(1, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      (bn0_1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running
_stats=True)
      (conv0_2): Conv2d(32, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      (bn0_2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running
_stats=True)
      (relu): ReLU(inplace=True)
      (maxpool1): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_m
ode=False)
      (layer1): Sequential(
        (0): BasicBlock(
          (conv1): Conv2d(64, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (relu): ReLU(inplace=True)
          (downsample): Sequential(
            (0): Conv2d(64, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          )
        )
      )
      (conv1): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
      (maxpool2): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_m
ode=False)
      (layer2): Sequential(
        (0): BasicBlock(
          (conv1): Conv2d(128, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (relu): ReLU(inplace=True)
          (downsample): Sequential(
            (0): Conv2d(128, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          )
        )
        (1): BasicBlock(
```

```
          (conv1): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (relu): ReLU(inplace=True)
        )
      )
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
      (maxpool3): MaxPool2d(kernel_size=2, stride=(2, 1), padding=(0, 1), dilation
=1, ceil_mode=False)
      (layer3): Sequential(
        (0): BasicBlock(
          (conv1): Conv2d(256, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (relu): ReLU(inplace=True)
          (downsample): Sequential(
            (0): Conv2d(256, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          )
        )
        (1): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (relu): ReLU(inplace=True)
        )
        (2): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (relu): ReLU(inplace=True)
        )
```

```
        (3): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (relu): ReLU(inplace=True)
        )
        (4): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (relu): ReLU(inplace=True)
        )
      )
      (conv3): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
      (layer4): Sequential(
        (0): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (relu): ReLU(inplace=True)
        )
        (1): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (relu): ReLU(inplace=True)
        )
        (2): BasicBlock(
          (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
```

```
1), bias=False)
          (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runn
ing_stats=True)
          (relu): ReLU(inplace=True)
        )
      )
      (conv4_1): Conv2d(512, 512, kernel_size=(2, 2), stride=(2, 1), padding=(0,
1), bias=False)
      (bn4_1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
      (conv4_2): Conv2d(512, 512, kernel_size=(2, 2), stride=(1, 1), bias=False)
      (bn4_2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
    )
  )
  (AdaptiveAvgPool): AdaptiveAvgPool2d(output_size=(None, 1))
  (SequenceModeling): Sequential(
    (0): BidirectionalLSTM(
      (rnn): LSTM(512, 256, batch_first=True, bidirectional=True)
      (linear): Linear(in_features=512, out_features=256, bias=True)
    )
    (1): BidirectionalLSTM(
      (rnn): LSTM(256, 256, batch_first=True, bidirectional=True)
      (linear): Linear(in_features=512, out_features=256, bias=True)
    )
  )
  (Prediction): Linear(in_features=256, out_features=79, bias=True)
)
```

In [63]:
```python
#Inputing Test Dataset to the model:
all_preds = [] #List to store predicted labels
all_labels = [] #List to store ground truth labels

model.eval()

with torch.no_grad(): #No gradients to be calculated
    for images, labels in tqdm(test_loader, desc="Testing"): #iterating through tes
        #images = images.to(device) #No need
        preds = model(images, None) #Model predicts lables from image data
        decoded_preds = greedy_decode(preds, idx_to_char) #greedy decoding
        all_preds.extend(decoded_preds)
        all_labels.extend(labels)
```

```
esting: 100%|████████████████████████████████████████████████████████████
████| 3/3 [00:10<00:00,  3.35s/it]
```

Printing Ground Truth labels and their corrosponding predictions for the complete test dataset:

In [64]:
```python
for pred,gt in zip(all_preds,all_labels):
    print(f"Ground Truth Label: {gt}")
    print(f"Label Predicted by OCR: {pred}")
    print()
```

Ground Truth Label: aunque con mayores ayudas que los Griegos, por tener mas conocid
os los manantiales de su lengua-
Label Predicted by OCR: dunque coninayores ayudas que los G.iegos, por tenermas cono
cidos los manantibles desu cnqua

Ground Truth Label: tadores hizieron la dicha particion con declara-
Label Predicted by OCR: tadores hizieron la dicha particion con declara-

Ground Truth Label: como un cuchillo solo se
Label Predicted by OCR: como ,n cuchitlo solose

Ground Truth Label: años de presidio, dos.
Label Predicted by OCR: años de presidio, dos,

Ground Truth Label: lla en 12. de Diziembre de 1637. ante Antonio de
Label Predicted by OCR: lla en l1 2. de Diziembre de 1637. ante Antonio de

Ground Truth Label: nor Rauasco, y por setenta mil marauedis para la
Label Predicted by OCR: nor Ravasco, y por se tenta mil marauedis para la

Ground Truth Label: recerles. Sean tan fervorosamen-
Label Predicted by OCR: recerles Sean tan fervorosamen-

Ground Truth Label: d leg. IuI. de adult. haec Padilla, Sua-
Label Predicted by OCR: a leg lul de adult hxe Padilla, sua-

Ground Truth Label: su. En las cuales manifiesta el trato ilícito que con él
Label Predicted by OCR: s Enlas quales manifiesta el trato ilicito que con el

Ground Truth Label: vinae, & humana domus, I. aduersus, C. de crimine ex-
Label Predicted by OCR: cuina, & humana domus, haduersus, C.de crimine ex-y

Ground Truth Label: es el fuego, cuerpo mas puro de quantos la taturaleza hizo; seri
a nunca acabar tratar esta materia.
Label Predicted by OCR: es el fiego y cuer po mas puro de quantos la tatura leza hiz
o: seria nuncaa cabar tratar ena mateada

Ground Truth Label: bando evidentemente lo contrario dellos; y enseñando lo uno que
Label Predicted by OCR: han io euideneemente lo contrario dellos, y ensciiando lo un
o que

Ground Truth Label: te con el criado que el cocinero ha dicho libremente que
Label Predicted by OCR: tes conel criado, que el vocinero ha duho libremente q

Ground Truth Label: CAR
Label Predicted by OCR: CAR

Ground Truth Label: siglo. Las pisadas destos tales tres varones impressas en este C
olegio, mueven mucho a los que en
Label Predicted by OCR: siglo. Las pisadas destos tales tres varones imprestas en es
te Colegio, mueven mucho a los que en

Ground Truth Label: testib us cum notatis ibidem per Canonitas, …Bart in
Label Predicted by OCR: testibus cum. notati5 ibi dem per Canonistas, … Bart in

Ground Truth Label: , porque el tenia pagado a

Label Predicted by OCR: porque el tenia pagado a

Ground Truth Label: tidades, se da por pagado de la dote, y bienes gana
Label Predicted by OCR: tidades, se da por pagado de la dote, y bienes gana

Ground Truth Label: cipio ni el primero pliego del, ni entregue
Label Predicted by OCR: cipio ni el primer pliego del, ni entre gue

Ground Truth Label: tantas, y tan grandes causas de estima: pero desta materia no es sujeto capaz una carta, y assi vengo
Label Predicted by OCR: rantas, y tan grandes causas de estima: pero desta materia n o es sujeto capaz una carta y asti vengo

Ground Truth Label: tio de v. m. fue el señor don Antonio de Covarruvias, hermano su yo, primero del Consejo Real
Label Predicted by OCR: tio de u. m. fue el señor Don Antonio de Covarruvias, herman o suyo, primero cel Consejo Rcal

Ground Truth Label: liberal con los Niños, pues no
Label Predicted by OCR: liberal con los Niños, pues no

Ground Truth Label: quam in cateris, sunt enim duo illi in carne una, c. gaude-
Label Predicted by OCR: quam in cnteris suntenim duo illincarne unale gande-

Ground Truth Label: o Niño tierno, y Dios Eterno,
Label Predicted by OCR: dNiño tierno, y Dios Eterno,

Ground Truth Label: de adult. in eis verbis: Vel per evidentiamrei, vel per
Label Predicted by OCR: de adult.in cis veibis Vel per euidentiamrei, vel per

Ground Truth Label: cablos, y la metafora que enriqueze todos los lenguages del mund o, y particularmente el nuestro,
Label Predicted by OCR: cablos, y la metafora que enriqueze todos los lenguages del mundo, y particularmente el cuestro,

Ground Truth Label: der
Label Predicted by OCR: der

Ground Truth Label: Estando condenado Iacinto Gomez en la
Label Predicted by OCR: Eestando condenado Iacinto Gomez en la

Ground Truth Label: diola monje professo de
Label Predicted by OCR: dio la monje professo de

Ground Truth Label: ta, con que los mil ducados fuessen ochocientos,
Label Predicted by OCR: ta, con que los mil ducados fuessen ochocientos,

Ground Truth Label: NIÑO JESUS.
Label Predicted by OCR: NINO ESUS

Ground Truth Label: gastado a su padre, y por ser menor de veinte y cin
Label Predicted by OCR: gastado a su padre, y por ser menor de veinte y cin

Ground Truth Label: DE
Label Predicted by OCR: DE

```
Ground Truth Label: quando mater esset vilis persona, uel suspecta de collu-
Label Predicted by OCR: quando marer este vilis persona vel suspecta de collu-

Ground Truth Label: varon, unico sucessor en el mayorazdo de su padre, mo-
Label Predicted by OCR: varon, unico sucessor en el mayorazgo de su padre, mo-

Ground Truth Label: Colegio de San Salvador de Oviedo, el mayor de Salamanca, Sacris
tan
Label Predicted by OCR: Colegio de San Saluador de Oviedo, el mayor de Salamanca, Sa
cristan

Ground Truth Label: cumentos, mereceran, quando hombres,
Label Predicted by OCR: cumentos, mereceran, quando hombres,

Ground Truth Label: consistit in veritate, & non infictione, & est focia vi-
Label Predicted by OCR: confistitin veritate, & non infictione, & est socia di-

Ground Truth Label: de los señores Alcaldes, en que man-
Label Predicted by OCR: de los señores Alcaldes, en que man-

Ground Truth Label: mucha
Label Predicted by OCR: mucha

Ground Truth Label: mos en la trampa. Bien sé que estas mis palabras se echa-
Label Predicted by OCR: mos en la trampa. BienSeque estas mis palabras se echa
```

As we can see above, the CRNN model is able to carry out OCR on the images with a good level of accuracy.However, some character to character differrences still remain in some cases. Calculating Error Metrics to quantify these differrences:

In [65]:
```
#Final evaluation metrics:
test_cer = calculate_cer(all_preds, all_labels)
test_wer = calculate_wer(all_preds, all_labels)

print("Test CER:", test_cer)
print("Test WER:", test_wer)
```

```
Test CER: 0.06680265170831208
Test WER: 0.33053221288515405
```

Thus, when the CRNN model is run on the **Test Dataset**, about **93% of characters are predicted correctly**.Thus, 6.6% of characters in the predicted text differ from the ground truth.

Conclusion:

An end-to-end OCR pipeline for historical Spanish printed text was implemented using DBNet for text detection and CRNN for text recognition. The trained model achieved a **CER of 6.6% on Test data and 5.9% on Validation data** demonstrating the effectiveness of the implemented framework.

Preliminary experiments with LLM-based post-processing were attempted. However, this did not result in significant improvements in CER within the current framework, indicating that the refinement strategy requires further redesign.

Future work will focus on the following directions:

1. Implementing beam search decoding in place of greedy decoding to improve sequence prediction.
2. Developing a more robust LLM-based post-processing framework for refining CRNN outputs.
3. Exploring alternative text recognition architectures beyond CRNN to further improve OCR accuracy.

In [66]:
```
!jupyter nbconvert --to webpdf OCR_on_Printed_Historical_Text_final_code.ipynb
```

```
[NbConvertApp] Converting notebook OCR_on_Printed_Historical_Text_final_code.ipynb t
o webpdf
[NbConvertApp] WARNING | Alternative text is missing on 6 image(s).
[NbConvertApp] Building PDF
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 1514269 bytes to OCR_on_Printed_Historical_Text_final_code.pd
f
```

In [ ]:
```
```